II श्री गणेश प्रसन्न II

CAMPBELL-WALSH-WEIN · BOOK 1

Basic Principles (General Urology)

Concise rapid revision · facts answered in Papers 1–4 point to that answer

highlighted number to memorise · KEY POINT and ALGORITHM boxes carry the reasoning · P2 = covered in full in that Paper's answer book; the list under it gives what that answer misses

Chapter	Topic
Ch 1	Personal and Professional Development of the Urologist
Ch 2	Evaluation of the Urologic Patient: History and Physical Examination
Ch 3	Indications and Interpretation of Laboratory Testing
Ch 4	Basic Principles of Urologic Ultrasonography
Ch 5	Principles of CT and Radiographic Imaging
Ch 6	Magnetic Resonance Imaging
Ch 7	Genitourinary Tract Imaging: Basic Principles of Nuclear Medicine
Ch 8	Evaluation and Management of Haematuria
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Ch 12	Fundamentals of Intraoperative Patient Safety
Ch 13	Incisions, Exposure and Drains
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Ch 15	Lower Urinary Tract Catheterisation
Ch 16	Fundamentals of Upper Urinary Tract Drainage
Ch 17	Principles of Urologic Endoscopy
Ch 18	Fundamentals of Laparoscopic and Robotic Urologic Surgery
Ch 19	Energy Modalities in Urology
Ch 20	Complications of Urologic Surgery
Ch 21	Urologic Considerations in Pregnancy
Ch 22	Intraoperative Consultations
Ch 23	Assessment of Urologic and Surgical Outcomes
★	Putting it together: everyday decisions

1 Personal and Professional Development of the Urologist

1 · Professionalism

- **Unprofessional behaviour in medical school** is one of the strongest predictors of later disciplinary action by medical boards.
- Surgeons in the **highest quartile of patient complaints have more surgical complications**, so unprofessional behaviour is a measurable safety signal.
- The Patient Advocacy Reporting System (PARS) uses patient complaints, and the Coworker Reporting System (CORS) uses co-worker reports. After a CORS report, most individuals are never reported again.

2 · Burnout

- **Definition (triad)**: emotional exhaustion, depersonalisation and diminished personal accomplishment, in response to chronic interpersonal stressors. It is distinct from depression.
- Prevalence is about **40%** of physicians, and urologists approach **50%**. It causes more medical error.
- Physician suicide is **1.4×** the population rate in men and **2.7×** in women.
- **20% rule**: spending less than 20% of effort on the most meaningful activity raises burnout risk about **3×**.
- Drivers are loss of autonomy, purpose and community, electronic health record time and administrative burden.

Prevention strategy	Effect
Institutional awareness seminars	Minimally effective
Practice interventions (scribes)	More effective
Restoring autonomy over schedule	Significantly reduces burnout
Mindfulness, communication training, exercise	Evidence-based reduction

> **KEY POINT**
> Interventions that restore autonomy, purpose or community work; awareness alone does not.

3 · Four domains of learning

Domain	Framework	Assessment
Knowledge	Bloom's taxonomy (apex: create)	Easiest; certification exams
Psychomotor	Imitation, manipulation, precision, articulation, naturalisation (novice to master)	Proctorship (often needed for robotic and laser credentialing), simulation
Communication	TeamSTEPPS	Often untaught
Affective	Krathwohl's taxonomy (apex: characterisation by a value set)	Formative

P4 Operating room safety: the role of surgeon — TeamSTEPPS tools (brief, debrief, SBAR, check-back, CUS, two-challenge rule)

Not in the paper answer:

- Communication errors contribute to about **70%** of medical errors and **27%** of malpractice claims.
- Situation awareness belongs to the individual, while a shared mental model belongs to the team.

> **KEY POINT**
> Certification tests the easiest domain, knowledge, but harm arises mainly in the communication, affective and psychomotor domains.

2 Evaluation of the Urologic Patient: History and Physical Examination

1 · Pain

- **Pain without tenderness is referred pain**: a non-tender painful testis may reflect an ipsilateral ureteric stone.

	Obstructive (colic)	Parenchymal
Cause	Distension of a hollow organ	Capsular stretch (pyelonephritis)
Pattern	Intermittent; restless, seeks a position	Constant; costovertebral angle tenderness; lies still

- Colic is mediated by prostaglandins and perceived in the **T11–L1** segments, which is why non-steroidal anti-inflammatory drugs (NSAIDs) work. A ureterovesical junction stone causes urgency and frequency.
- **Strangury** is stabbing pain at the end of micturition from an inflamed detrusor. An atonic bladder retains large volumes painlessly.

2 · Haematuria

P2 P3 Microscopic Hematuria; Asymptomatic haematuria and urinary biomarkers — ≥3 RBC/HPF, AUA/SUFU 2020 risk groups and work-up, anticoagulants do not explain it

Not in the paper answer:

- The commonest cause of gross haematuria over 50 years is **bladder cancer**.

Clue	Suggests
Initial haematuria	Urethra or prostate
Total haematuria	Bladder or above
Terminal haematuria	Bladder neck
Vermiform clots / cuboid clots	Upper tract / bladder
Clears with irrigation but recurs / clears once a catheter is placed	Upper tract / prostate

3 · Lower urinary tract symptoms

- A **bladder diary** (usually 48 h) gives functional capacity (normal **300–400 mL**), nocturia and polyuria. Frequency means more than 5–6 voids a day.
- Daytime frequency without nocturia suggests a psychogenic cause. Nocturia without daytime frequency suggests nocturnal polyuria.
- Irritative symptoms may be the only sign of carcinoma in situ (CIS), so keep a low threshold for cystoscopy in smokers over 50.

P3 IPSS score — 7 items scored 0–5 (total 0–35; mild 0–7, moderate 8–19, severe 20–35) and a separate quality-of-life item scored 0–6

4 · Urinary incontinence

Type	Mechanism	Hallmarks
Continuous	Outlet bypassed (fistula, ectopic ureter)	Constant wetness

Type	Mechanism	Hallmarks
Overflow	Volume exceeds capacity	Often nocturnal; long-standing obstruction; cured by relieving it
Stress	Sphincter weakness	Cough or sneeze; multiparity; after prostatectomy
Urge	Leakage with urgency	Overactive bladder, cystitis, occult cancer
Mixed	Stress plus urge	Treating one alone may worsen symptoms
Functional	Cannot reach the toilet	Mobility, cognition

P3 Embryology and management of ectopic ureter — a girl who dribbles continuously with normal voiding
P1 P3 Nocturnal enuresis; Enuresis — definition, alarm and desmopressin

Not in the paper answer:

- Enuresis affects about **15%** at 5 years and **1%** at 15 years. Primary enuresis beyond 6 years needs urological evaluation, and secondary enuresis should raise the possibility of abuse or bullying.

5 · Sexual and other symptoms

- **Erectile dysfunction (ED)** (National Institutes of Health 1993) is inability to attain or maintain an erection sufficient for satisfactory intercourse. **ED heralds occult coronary artery disease.**
- With low libido and ED, measure early-morning testosterone on two occasions **≥2 weeks** apart.

P1 P3 Premature Ejaculation — ISSM definitions (lifelong about 1 min, acquired ≤3 min) and treatment
P1 Haematospermia — usually benign; when to investigate

Not in the paper answer:

- The AUA/Sexual Medicine Society definition of premature ejaculation is within about 2 min.
- The commonest cause of **pneumaturia** is an enterovesical fistula (diverticulitis, Crohn disease). Bloody urethral discharge suggests urethral carcinoma.

6 · Drug history

Side effect	Drugs
ED	Thiazides, propranolol
Ejaculatory dysfunction	α-blockers, methyldopa, phenothiazines
Priapism	**Trazodone**, phenothiazines, hydralazine, prazosin
Retention	Anticholinergics, calcium-channel blockers, α-agonists (pseudoephedrine), antihistamines
Reduced spermatogenesis	Alkylating agents, marijuana
Gynaecomastia	Digoxin, cimetidine, verapamil

Antithrombotic	Action	Half-life
Warfarin	Vitamin K antagonist	36–42 h
Dabigatran	Direct thrombin inhibitor	8–17 h
Rivaroxaban / apixaban	Factor Xa inhibitors	5–12 h / 12 h

Antithrombotic	Action	Half-life
Clopidogrel / ticagrelor	P2Y12 blockade, irreversible / reversible	6 h / 6–8 h

- Half-life is not duration: **clopidogrel acts for the life of the platelet**.
- **Bleeding risk** is low for cystoscopy and ureteroscopy, intermediate for transurethral resection of the prostate (TURP) and of bladder tumour (TURBT), and **high for percutaneous nephrolithotomy (PCNL)**, artificial urinary sphincter and robotic nephrectomy.

7 · Physical examination

- **Bladder**: palpable above about **150 mL** and visible at about 500 mL. **Bimanual examination under anaesthesia after TURBT** is standard of care and independently predicts pT3.
- **Prostate**: a nodule as hard as a flexed knuckle needs biopsy.
- **Torsion**: horizontal lie, high testis and short cord with pain. An absent cremasteric reflex is highly sensitive in children; a present reflex argues against torsion but does not exclude it.
- Normal testis is **4–5 × 2–3 cm**. A **hard testicular mass is cancer until proven otherwise**, whereas an epididymal mass is almost always benign.
- **Female**: a Q-tip deflection **≥30°** on Valsalva means urethral hypermobility, but it is not required to diagnose stress incontinence. Document prolapse by POP-Q.
- **Neuro-urology**: the bulbocavernosus reflex tests **S2–S4**. Saddle sensory loss with an absent reflex localises a sacral root lesion such as cauda equina.

3 Indications and Interpretation of Laboratory Testing

1 · Urinalysis: collection and gross examination

- Examine urine within **2 h** or refrigerate it at 2–8 °C. Refrigeration precipitates crystals.
- In children, suprapubic aspiration has the lowest contamination and catheterisation is often preferred. A **bag specimen is never used for culture**.
- Red urine is not always blood (beetroot, rifampicin, phenothiazines).
- Cloudy urine is most often **phosphaturia**, which clears with acetic acid; pyuria does not.

2 · Dipstick chemistry

Interferent	False negative	False positive
Ascorbic acid	Glucose, bilirubin, blood, leucocyte esterase	–
Phenazopyridine / dyes	Leucocyte esterase	Nitrite, bilirubin, ketones
Alkaline urine	–	Protein
High specific gravity	Leucocyte esterase, glucose	–
Semen, menses, exercise	–	Blood
Mesna, levodopa	–	Ketones
Contamination (commonest)	–	Leucocyte esterase and nitrite

- **Specific gravity** is 1.001–1.035 and becomes **fixed at 1.010 in renal insufficiency**.
- pH is 4.5–8.0. A pH **>7.5** with infection means a urease-splitting organism (Proteus), which leads to struvite staghorn stones. Uric acid and cystine stones form in acid urine.

P1 P2 P3 Distal Renal tubular acidosis; Renal tubular acidosis; Type 2 Renal tubular acidosis — cannot acidify below pH 5.5; calcium phosphate stones

- The renal threshold for glucose is a serum level of about **180 mg/dL**, and SGLT2 inhibitors cause glycosuria. Nitroprusside detects acetoacetate but **not β-hydroxybutyrate**.

3 · Haematuria

P2 P3 Microscopic Hematuria; Asymptomatic haematuria and urinary biomarkers — ≥3 RBC/HPF; confirm a dipstick by microscopy

Not in the paper answer:

Red urine	Sediment	Supernatant	Plasma
Haematuria	RBCs	Clear	–
Haemoglobinuria	No RBCs	Red, haem positive	Red
Myoglobinuria	No RBCs	Red, haem positive	Clear
Dyes, porphyria	No RBCs	Red, haem negative	–

Plasma is red in haemoglobinuria because haemoglobin is haptoglobin-bound; small unbound myoglobin is filtered at once.

	Glomerular	Non-glomerular medical	Urological
RBC shape	Dysmorphic	Round	Round
RBC casts	Present	Absent	Absent

	Glomerular	Non-glomerular medical	Urological
Proteinuria	Significant	Often significant	Minimal, even with gross bleeding

- Proteinuria of 2+–3+ (100–300 mg/dL) is almost never urological, so refer to nephrology.

P1 Berger's Disease — IgA nephropathy, the commonest glomerular cause; haematuria after an upper respiratory infection

- Other clues are deafness with a family history (Alport syndrome), haemoptysis (Goodpasture syndrome), and a recent throat infection with low C3 (poststreptococcal glomerulonephritis).

4 · Proteinuria

P1 Proteinuria — glomerular, tubular and overflow types; dipstick misses light chains (use sulfosalicylic acid)

5 · Infection: leucocyte esterase, nitrite and culture

	Leucocyte esterase	Nitrite
Detects	Host response (neutrophils)	Organism (gram-negative nitrate reductase)
Performance	Sensitive, non-specific	Specific, insensitive
Classic miss	Glycosuria, proteinuria	**Enterococcus**; low counts

- If only one is positive, do microscopy and culture before antibiotics. A dipstick never replaces culture.
- Culture thresholds are **≥10^5 CFU/mL** for clean catch and **≥10^3 CFU/mL** for a catheter specimen; **any growth on suprapubic aspiration is significant**.

6 · Sediment microscopy

- Squamous cells mean contamination, while **renal tubular cells always mean renal pathology**. Eosinophils suggest acute interstitial nephritis.
- Schistosoma haematobium ova have a **terminal spike**.

Cast	Meaning
Hyaline	May be normal (exercise, heat)
RBC	Glomerulonephritis
WBC	Pyelonephritis, interstitial nephritis
Muddy-brown granular	Acute tubular necrosis
Fatty	Nephrotic syndrome

P2 Urinary crystals — cystine hexagons always pathological; struvite coffin lids

7 · Serum tests

- Check creatinine and estimated glomerular filtration rate (eGFR) whenever obstruction is suspected, because renal function sets the urgency of drainage.
- Creatinine rises only after about **50%** loss of GFR. CKD stages by eGFR (mL/min/1.73 m^2) are G1 ≥90, G2 60–89, G3a 45–59, G3b 30–44, G4 15–29 and G5 <15.

8 · Tumour markers and cytology

P1 Tumour markers in urology — PSA organ-specific, not cancer-specific; germ cell markers and half-lives

P1 Urine cytology — specific for high grade and CIS; not for screening or initial microhaematuria

9 · Stone, adrenal and andrology tests

P1 P2 Metabolic Evaluation in Stone Disease — screen every first-time former; 24-h urine if high risk

P2 Algorithm for management of an incidentaloma — dexamethasone suppression, metanephrines, aldosterone-to-renin ratio

Not in the paper answer:

- Aldosterone **>15 ng/dL** with an aldosterone-to-renin ratio **>20** leads to adrenal vein sampling.

P4 Semen analysis nomenclature — WHO 2021 limits (1.4 mL, 16 million/mL, 39 million, 42%/30%, 4%, 54%), abstinence 2–7 days

Not in the paper answer:

- Leucocytes **>1 million/mL** define leucocytospermia.

10 · Office tests

P2 P3 Uroflowmetry — Qmax is the key value; <10 mL/s suggests obstruction but cannot exclude underactivity

Not in the paper answer:

- Qmax **>15 mL/s** gives a low probability of obstruction and 10–15 mL/s is equivocal. A post-void residual **<100 mL** is generally acceptable.
- **No antibiotic prophylaxis** for simple cystoscopy or urodynamics without risk factors (age, steroids, immunodeficiency, anatomical anomaly, colonised material, long hospitalisation).

4 Basic Principles of Urologic Ultrasonography

1 · Physical principles

- Piezoelectric crystals convert alternating current into a longitudinal mechanical wave. Echo amplitude gives pixel brightness, and round-trip time gives depth. The urological range is **3.5–20 MHz**.
- Velocity equals frequency × wavelength. Speed of sound in soft tissue is **1540 m/s**, so frequency and wavelength vary inversely.

	Axial resolution	Lateral resolution
Separates	Objects one behind the other	Objects side by side
Improved by	Higher frequency	Narrow beam: focal zone at the depth of interest

KEY POINT

Absorption is proportional to frequency, so high frequency resolves finely but penetrates poorly: **7–18 MHz** for the testis and **3.5–5 MHz** for the adult kidney.

- **Reflection depends on the impedance difference at an interface, not on density.** Soft tissues cluster together (1.4–1.7 × 10^6 kg/m²·s), while air (0.0004) and bone (7.8) are extremes. Hence gel, bowel-gas obscuration and stone shadowing.

Artefacts

Artefact	Mechanism	Appearance
Acoustic shadowing	Strong reflection or attenuation	Dark band behind a stone; the stone looks crescentic, so size is inaccurate
Increased through-transmission	Less attenuation than surroundings	Bright tissue behind a simple cyst
Edging (critical angle)	Refraction along a curved surface	Shadow at the testicular pole
Reverberation	Bounce between two strong reflectors	Equally spaced echoes; the second at **twice the depth**
Comet tail	Many small reflectors (bowel gas)	Bright tail; may hide the lower renal pole
Twinkling	Rough reflective surface on colour Doppler	Colour mosaic behind a stone; detects small non-shadowing stones

2 · Modes

P1 Doppler in urology — resistive index (RI); colour, spectral and power (3–5× more sensitive) modes; organ findings

Not in the paper answer:

- **Elastography**: strain elastography (manual compression) is qualitative and operator-dependent. **Shear-wave elastography is quantitative** (shear waves 1–10 m/s; $E = 3\rho c^2$). Hard does not mean malignant.

3 · Contrast-enhanced ultrasound (CEUS)

P1 Non iodinated contrast materials — microbubbles are exhaled and not nephrotoxic; uses in renal masses and voiding urosonography

Not in the paper answer:

- Bubbles cannot cross the glomerulus and **never opacify the collecting system**.

4 · Safety and ALARA

	Mechanical	Thermal
Nature	Cavitation, streaming	Heating
Greatest	Near gas: lung, bowel	High frequency, focusing, long dwell, Doppler
Index	Mechanical index (MI): **≤0.7** low risk; limit time if >0.4 next to lung or bowel	Thermal index (TI) is the probability of a 1 °C rise; keep **<2** in urology

- **MI and TI are displayed estimates, not safety limits.**
- ALARA (as low as reasonably achievable) means low power, short examinations and adjusted focus. The **informed operator is the most important safety factor**.

5 · Renal ultrasound

- Normal kidney length is **10–12 cm** and width 4–5 cm, measured in the true midsagittal plane. Cortex is **>7 mm**, parenchyma **>15 mm** and intrarenal RI **0.47–0.70**.
- Cortex is slightly hypoechoic to liver; an iso- or hyperechoic cortex suggests medical renal disease.
- **Pyramids mimic calyces**: pyramids lie outside the hilar fat, and the collecting system lies inside it.
- Ultrasound-guided PCNL access is hard with a long skin-to-stone distance, a non-dilated system or a staghorn.
- After renal biopsy, complications rise with **severe hypertension, not the number of passes**. Haematuria appears within 6 h, so 6 h of observation equals admission.

P4 Radiology for solid renal masses — grey-scale misses masses <2 cm; a hyperechoic mass may be angiomyolipoma or renal cell carcinoma, so get CT

6 · Bladder and pelvic ultrasound

- Bladder volume = W × H × L × **0.625**.

P1 Methods to estimate prostate size — ellipsoid volume = W × H × L × 0.52

- **Bladder wall thickness predicts bladder outlet obstruction better than uroflowmetry**, post-void residual or prostate volume.
- **An automated bladder scanner is not an imaging study.** Fluid outside the bladder (ascites, urinoma, lymphocele) gives a wrong residual.

7 · Scrotal ultrasound

- Show both testes side by side, because **diffuse lymphoma or leukaemia is visible only by comparison**.
- Intratesticular PSV is **<10 cm/s**, and an RI **>0.6** suggests impaired spermatogenesis.

KEY POINT

In torsion, epididymal collaterals can appear within hours, and reactive hyperaemia can give a false-negative study. Ultrasound cannot diagnose torsion; only the surgeon can, so strong suspicion means exploration.

8 · Penile and urethral ultrasound

P1 Doppler in urology — penile duplex after intracavernosal injection: PSV ≥35 normal, 25–35 equivocal, <25 cm/s arteriogenic; EDV >5 cm/s with RI <0.75 means veno-occlusive dysfunction

Not in the paper answer:

- Arterial dilation **<75%** and acceleration time **>110 ms** also indicate arteriogenic ED. PSV is the most accurate measure, and arteriogenic ED flags systemic cardiovascular disease.

9 · Transrectal ultrasound (TRUS) and therapy

P1 P2 Transrectal ultrasound study of prostate; TRUS — volume, PSA density, micro-ultrasound (29 MHz) comparable to MRI

P2 HIFU — focused thermal ablation of prostate cancer; late rectourethral fistula

- **Pregnancy**: ultrasound is first-line and safe in all trimesters, but grey-scale is poor for ureteric stones.

5 Principles of CT and Radiographic Imaging

1 · Radiation biology and safety

	Deterministic	Stochastic
Threshold	Yes	**None** (linear no-threshold model)
Dose increases	Severity	Probability (cumulative)
Mechanism	Cell loss	Mutation in an oncogene or tumour suppressor
Examples	Erythema, epilation, sterility, cataract	Cancer

- Absorbed dose is in gray (Gy). **Effective dose** (sievert, Sv) adds tissue weighting and sets exposure limits.

Study	Effective dose
Chest x-ray	**0.07 mSv**
Kidney–ureter–bladder (KUB) film / two-view abdomen	**0.7 / 1.4 mSv**
Ultra-low-dose CT abdomen–pelvis	About 1 mSv (1–2)
Low-dose CT	<3–4 mSv
Conventional CT abdomen–pelvis	**10–20 mSv**; 10 mSv carries about a 1 in 1000 cancer risk

KEY POINT

Stochastic risk has no threshold, so no dose is safe. Hence ALARA (as low as reasonably achievable): a risk–benefit decision before every ionising study, low-dose protocols and non-ionising alternatives.

2 · Radiographic studies

- KUB is used to follow radio-opaque stones, before shock-wave lithotripsy. CT detects more than **95%** of stones. An **upright film is most sensitive for free air**.

P1 Describe IVU & its applications in Urology — now largely replaced by CT urography

P2 Lower urinary tract trauma in pelvic fracture — plain and CT cystography (retrograde 350 mL), flame-shaped extraperitoneal versus intraperitoneal leak

Not in the paper answer:

- Passive filling by excreted contrast misses ruptures. Extraperitoneal rupture can give a "molar tooth" pattern.
- A large extraperitoneal leak prevents distension and can **hide a coexisting intraperitoneal rupture**.

3 · Fluoroscopic studies

Setting	Peak kilovoltage (kVp)	Tube current (mA)
Changes	Photon energy	Photon number
Dose	**Rises exponentially**	Rises linearly
Penetration / contrast	Increases / decreases	Unchanged / increases

- **Retrograde pyelography** is chosen when intravenous contrast is hazardous (allergy, renal dysfunction). Use about 50% contrast, because dense contrast hides small filling defects. Over-pressure causes pyelovenous and pyelolymphatic backflow and sepsis.

P1 VCUG — gold standard for reflux; capacity (age × 30) + 30 mL for ages 2–14

- **Retrograde urethrography (RUG)** shows the anterior urethra. Use a 45° oblique, the balloon at about 2 mL in the fossa navicularis, and 20–30 mL of contrast. The prostatic urethra does not distend normally, which is not a stricture; the posterior urethra needs a combined VCUG.
- **Fluoroscopic renal access**: bull's-eye uses a C-arm tilted 15–30° toward the surgeon, with the needle end-on over the calyx.

4 · Fluoroscopy dose reduction

- Dose is proportional to fluoroscopy time. Recording it reduces it.
- Halving the collimator diameter cuts the exposed area **fourfold**. Low-dose mode halves mA and dose. Pulsed fluoroscopy at 8, 4 and 1 pulses/s saves **73%, 87% and 97%** against continuous (30 pulses/s).
- **Inverse square law**: doubling the distance gives ¼ of the dose.
- Place the source below the patient and the detector close to the patient. Wear a **0.25–0.5 mm** lead apron, a thyroid shield and glasses; the legs receive most scatter.

> **KEY POINT**
> Surgeon dose is mostly scatter from the patient, so everything that lowers patient dose protects the team.

5 · CT principles

P1 Newer generation CT scan — dual-energy CT, dose protocols, contrast phases, enhancement ≥20 HU, stone HU

Not in the paper answer:

Material	HU
Macroscopic fat (diagnostic of angiomyolipoma, AML)	≤ −10
Simple cyst	−9 to 20
Lipid-rich adrenal adenoma	<10
Hyperdense cyst (Bosniak II)	≥70 unenhanced

- Partial-volume **pseudoenhancement of cysts <10–15 mm** leads to overtreatment, so use thin slices and a central region of interest.

6 · Noncontrast CT (NCCT) for stones

- NCCT is the gold standard; all stones are visible except **protease-inhibitor (indinavir) stones**. Cutting tube current by up to 95% preserves accuracy, and ultra-low-dose CT gives less dose than a two-view KUB.
- **Avoid ultra-low dose** with BMI >30, very thin patients (phlebolith confusion), metal hardware or a suspected infected stone.
- CT, low-dose and ultra-low-dose CT are equally accurate (sensitivity about 95%). Ultrasound has sensitivity **84%** but specificity only **53%**; KUB sensitivity is **57%**.

Body	First imaging in suspected colic
AUA	CT in adults; standard dose if BMI >30; NCCT before PCNL or SWL
EAU	Ultrasound first, CT if inconclusive (randomised trial: less radiation, same outcomes)

7 · Contrast CT protocols

- The portal venous phase (70–90 s) shows liver and nodes for routine staging. The nephrographic phase (90–120 s) is **most sensitive for renal masses** and caval thrombus. Clear cell carcinoma enhances early and avidly; papillary enhances later.
- **Adrenal protocol** is noncontrast, portal venous and 15-min delay. Absolute washout = (PV − delayed)/(PV − unenhanced) × 100 > **60%**, and relative washout = (PV − delayed)/PV × 100 > **40%**, indicate adenoma.
- Nodal cut-offs are 10 mm for prostate cancer (poor sensitivity, so use PSMA PET-CT) and, in bladder cancer, a short axis >8 mm in the pelvis and >10 mm in the abdomen.

P4 Radiology for solid renal masses — Bosniak 2019 classes, malignancy rates and IIF follow-up

8 · CT urography and CT cystography

- CT urography always includes an excretory phase. For poor distal ureteric opacification, **hydration is the simplest and most effective aid**, followed by a diuretic or prone position.
- Do **CT before TURBT**, because post-resection change mimics invasion. CT detects T3b/T4 but cannot separate muscle-invasive from non-invasive disease.

P1 P4 PET scans in urology; PSMA PET Ct scan — PSMA PET-CT detects recurrence even at PSA <0.5 ng/mL

9 · Iodinated contrast reactions and contrast-associated AKI

P1 Iodinated and other contrast agents (use in urology) and their complications; Acetylcysteine and Contrast Induced Nephropathy (CIN) — reactions, adrenaline dose, premedication, KDIGO AKI, saline for eGFR <30

> **KEY POINT**
> Hypersensitivity is idiosyncratic (premedication, adrenaline), whereas nephrotoxicity is dose-related (hydration, less contrast). Prophylaxis for one does nothing for the other.

6 Magnetic Resonance Imaging

1 · Physics and sequences

- Magnetic resonance imaging (MRI) gives **better soft-tissue contrast and no ionising radiation**.
- **Fluid is dark on T1 and bright on T2.**
- **Diffusion-weighted imaging (DWI)**: high cellularity restricts water movement, so tumour is bright on DWI and dark on the apparent diffusion coefficient (ADC) map.
- **Chemical shift imaging**: a voxel holding both fat and water loses signal on the out-of-phase (OP) image, which means **microscopic fat**. Fat suppression nulls **macroscopic fat**.

2 · Indications, contrast and safety

P1 Non iodinated contrast materials; MR urography — gadolinium chelation, nephrogenic systemic fibrosis (NSF) at glomerular filtration rate <30, American College of Radiology group II agents safe even in dialysis

Not in the paper answer:

- MRI is the modality of choice in **pregnancy, renal insufficiency and iodinated contrast allergy**.
- **Contraindications**: non-conditional pacemakers, ferromagnetic aneurysm clips, cochlear implants and orbital metal.

3 · Adrenal MRI

P2 P4 Algorithm for management of an incidentaloma; Adrenal mass evaluation — computed tomography (CT) density, washout, chemical-shift MRI

Not in the paper answer:

- An adenoma shows an **OP signal drop** from intracytoplasmic lipid. Lipid-poor adenomas (**10–30%**) are indeterminate and need CT washout.
- **Myelolipoma** has macroscopic fat: bright on T1, it loses signal with fat suppression and shows the India ink artefact.
- **Traps**: metastases from clear cell renal cell carcinoma (RCC), hepatocellular carcinoma or liposarcoma contain lipid and mimic adenoma. One-third of phaeochromocytomas are T2-dark, so the "light bulb" sign is unreliable.

4 · Renal mass

P4 Radiology for solid renal masses — MRI subtyping table, enhancement ≥15–20%

Not in the paper answer:

- An **absent pseudocapsule** predicts **pT3a**.
- Calcium is a signal void, so MRI shows enhancing nodules in **calcified cysts** better than CT. Tumour thrombus enhances; bland thrombus does not.

5 · MR urography and bladder

P1 P4 MR urography; VI-RADS — static T2 vs excretory T1; muscle-invasion score

Not in the paper answer:

- Upper tract urothelial carcinoma is homogeneous on DWI (ADC **≤451**), unlike central RCC.

- **T2 MRI is the best study for a urethral diverticulum**: it defines the neck, circumferential extent and any tumour or stone.

6 · Prostate MRI

P1 P4 Prostate multiparametric MRI and endorectal coil MRI; PI-RADS and MRI–ultrasound fusion biopsy — sequences, PI-RADS v2.1, biparametric MRI

Not in the paper answer:

- Prostate MRI has sensitivity **0.91** and specificity **0.37** for Grade Group ≥2, with negative predictive value **>90%**: it rules cancer out, not in.
- Dynamic contrast curves: type 1 persistent rise is normal, type 2 plateau is indeterminate, and **type 3 washout** is most suggestive.

Score	PI-QUAL (quality)	PRECISE (active surveillance)
1	All sequences non-diagnostic	Resolution
2	One sequence acceptable	Reduction
3	Rules in, not out	Stable
4	Rules in and out	Increase
5	All optimal	Stage progression

At PI-QUAL 1–2, give no PI-RADS score.

Trial	Design	Result	Take-home
PROMIS	MRI vs template biopsy	MRI highly sensitive	MRI before biopsy (level 1b)
PRECISION	Randomised: MRI ± targeted vs systematic biopsy	More significant, less insignificant cancer	MRI pathway standard
Göteborg-2	Screening trial	Targeted-only biopsy cut overdiagnosis	MRI-targeted screening

7 Genitourinary Tract Imaging: Basic Principles of Nuclear Medicine

1 · Principles and radionuclides

P1 Radionuclide (isotope) scans in urology: principles, renal scans, MAG3 and applications — gamma camera chain, planar vs single-photon emission computed tomography (SPECT) vs positron emission tomography (PET), ^{99m}Tc 6 h and 140 keV, ^{18}F sharpest image, ^{11}C needs a cyclotron

Not in the paper answer:

- **Gallium-68 (^{68}Ga)** has a half-life of **68 min**, is generator-produced and has the **longest positron range**.
- Indium-111 (half-life **67 h**) and gallium-67 (**78 h**) suit delayed infection imaging. Gallium-67 citrate is transferrin-bound and non-specific, imaged at 48–72 h.
- Renal blood flow is about **1 L/min** (20% of cardiac output), and renal plasma flow about **600 mL/min**.

2 · Renal tracers and diuretic renography

P1 P2 Radionuclide scans LAQ (as above); Radionuclide studies and diuretic renography in PUJ obstruction; Renal isotope scans and DMSA renal scan — DTPA, MAG3 and DMSA table, split function at 1–3 min (normal to 45:55), furosemide 0.5 mg/kg, half-clearance time <10 / 10–20 / >20 min, false-positive pitfalls, captopril (≥10% improvement), transplant rejection vs delayed graft function, DMSA scar persisting at 6 months, labelled leucocytes

Not in the paper answer:

- Always check **whether and when furosemide was given**. With F−15, furosemide goes 15 min before the tracer, so diuresis is maximal when the tracer arrives.
- Captopril renography has largely been replaced by computed tomography angiography and Doppler.

ALGORITHM

1. Day 1: give oral captopril and perform a renogram; slow uptake and a low peak are abnormal.
2. If abnormal, repeat 1–2 days later off angiotensin-converting enzyme (ACE) inhibitors and calcium channel blockers.
3. Improvement of **≥10%** means a high probability of renovascular hypertension and likely benefit from revascularisation.

3 · Oncological PET, bone scan and PSMA

P1 P4 PET scans in urology; PSMA PET Ct scan — fluorodeoxyglucose (FDG) for residual seminoma >3 cm but not teratoma, poor in prostate and clear cell renal cell carcinoma; approved prostate-specific membrane antigen (PSMA) ligands, nodal sensitivity 30–40% with specificity >90%, sestamibi vs girentuximab, Calais, CONDOR, VISION

Not in the paper answer:

- **Bone scan** (^{99m}Tc-methylene diphosphonate) binds hydroxyapatite at osteoblastic activity. It is non-specific, shows a **flare** early in therapy, and a **superscan** is diffuse uptake with **absent renal activity**.

- Choline and fluciclovine are taken up by inflammation and benign prostatic hyperplasia, so they perform poorly in the prostate itself. PSMA shows normal uptake in the salivary and lacrimal glands.
- The National Comprehensive Cancer Network does not require conventional imaging before PSMA-PET.

Trial	Design	Result	Take-home
Calais	^{68}Ga-PSMA vs fluciclovine, biochemical recurrence	Detection 56% vs 26%	PSMA superior
CONDOR	^{18}F-DCFPyL, biochemical recurrence	Detection 61.5% at prostate-specific antigen (PSA) 0.8 ng/mL; management changed 64%	Detects at low PSA
EMPIRE-1	Randomised: fluciclovine-guided vs conventional salvage radiotherapy	12.5% absolute 3-year event-free gain	PET-guided planning helps
VISION	Randomised, metastatic castration-resistant, after androgen receptor inhibitor and taxane	^{177}Lu-PSMA-617 approved	Radioligand therapy

8 Evaluation and Management of Haematuria

1 · Definitions and causes

P2 P3 Microscopic Hematuria; Asymptomatic haematuria and urinary biomarkers — microhaematuria (MH) is ≥3 red blood cells per high-power field (RBCs/HPF) on one microscopic urinalysis; dipstick false positives; prevalence 6.5%; malignancy ~3% (>85% bladder); glomerular picture needs nephrology plus urological evaluation; anticoagulants do not explain haematuria

Not in the paper answer:

- One positive urinalysis suffices because bleeding from cancer is intermittent. Specificity comes from **risk-stratifying the work-up, not from raising the threshold**.
- **Women** (male:female 4:1) present at a higher stage, because MH is put down to infection or gynaecological causes.
- The timing of gross haematuria (GH) in the stream points to the source:

Timing	Usual source
Initial	Urethra, distal to the external sphincter
Terminal	Trigone, bladder neck, prostate
Total	Bladder or upper tract
Cyclic (female)	Endometriosis

2 · Evaluation of microhaematuria (AUA/SUFU 2020)

P2 P3 As above — American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction (AUA/SUFU) risk table (low 0–0.4%, intermediate 1–2.5%, high ~6%), stratum-matched work-up, no cytology or markers initially, magnetic resonance urography if computed tomography (CT) is contraindicated, von Hippel–Lindau needs upper-tract imaging, repeat urinalysis within 12 months after a negative evaluation

Not in the paper answer:

- **Renal ultrasound** detects cortical tumours but is poorly sensitive for upper tract urothelial carcinoma; this is why CT urography is reserved for high risk.
- **CT urography** phases: non-contrast for stones, nephrographic for renal masses, excretory for urothelium. The dose is **20–30 mSv**, which a split-bolus protocol reduces.
- UroVysion fluorescence in situ hybridisation targets chromosomes **3, 7, 17 and 9p21**.

3 · Gross haematuria

- The cancer yield is several-fold higher than in MH, so there is **no age threshold and no stratification**.
- GH without trauma or culture-proven infection needs **cytology, cystoscopy and CT urography**. With culture-proven infection, treat it and then repeat the urinalysis.
- In acute bleeding, assess haemodynamics, blood count, coagulation and post-void residual.
- **Clot retention**: pass a large-bore (22–24 Fr) three-way catheter, evacuate all clot manually, then run continuous bladder irrigation (CBI) until clear. Evacuation must be complete, because residual clot keeps the bladder distended so vessels cannot contract, and it fuels fibrinolysis.
- If this fails, perform cystoscopy under anaesthesia with clot evacuation, fulguration and biopsy of suspicious areas.

- Decide on stopping or reversing anticoagulation with the prescriber, weighing thrombotic risk. Never blame the anticoagulant without a full evaluation.

4 · Haemorrhagic cystitis

P1 P3 Diagnosis and management of hemorrhagic cystitis; Management of Haemorrhagic cystitis — causes (acrolein, radiation endarteritis, BK virus), exclude cancer, mesna, escalation ladder from alum 1% to hyperbaric oxygen, formalin 1–4% after cystogram, anterior-division embolisation, cystectomy; RICH-ART

Not in the paper answer:

- **Silver nitrate 0.5–1%** also needs a cystogram first, because it precipitates in the upper tract.
- **Aminocaproic acid** inhibits plasminogen activation (urokinase) and so blocks fibrinolysis; use it only in a clot-free bladder.
- **Formalin technique**: instil by gravity with the catheter **≤15 cm** above the pubis, up to **300 mL** or bladder capacity, and protect the urethra and skin.

5 · Prostatic haematuria

- It is a diagnosis of exclusion after a full GH work-up. **Benign prostatic hyperplasia is the commonest cause of GH in men over 60**, through increased microvessel density driven by vascular endothelial growth factor (VEGF).
- **5α-reductase inhibitors** (finasteride, dutasteride) lower VEGF and microvessel density. About **90%** improve, including anticoagulated men, with onset from 2 weeks to 9 months. Randomised trials show fewer recurrences than with watchful waiting.
- Prostate cancer bleeding usually means T4 invasion. Treat with androgen deprivation therapy with or without palliative radiotherapy.

> **ALGORITHM**
>
> 1. Benign: start with a 5α-reductase inhibitor, then fulguration, then transurethral resection, laser vaporisation or enucleation, or simple prostatectomy, and finally prostatic artery embolisation (for poor-risk patients or bleeding after resection).
> 2. Malignant: start with androgen deprivation with or without radiotherapy, then fulguration or channel resection, then nephrostomy, and finally cystoprostatectomy.

6 · Urethral and upper tract bleeding

- **Urethrorrhagia** (blood at the meatus independent of voiding) means a lesion distal to the sphincter. Use retrograde urethrography and cystourethroscopy. With suspected trauma or penile fracture, **perform a urethrogram before any instrumentation**. In women, consider caruncle or diverticulum.
- **Renal arteriovenous malformation** is ~**75%** acquired (biopsy, partial nephrectomy, percutaneous nephrolithotomy, trauma). GH after a renal procedure needs prompt angiography and selective embolisation.
- **Ureteroiliac artery fistula** (radiation, vascular surgery, chronic stents) is treated by endovascular stenting.
- **Nutcracker syndrome**: the left renal vein is compressed between the aorta and the superior mesenteric artery.
- **Lateralising essential haematuria**: cystoscopy during bleeding shows the side, then ureteropyeloscopy inspects every calyx at low pressure for biopsy or fulguration.

Trial	Design	Result	Take-home
DETECT I	Prospective, UK, 3,556 referrals	Cancer in 13.8% with GH vs 3.1% with MH; 3.5% in GH under 45	Investigate GH at any age
RICH-ART	Randomised hyperbaric oxygen vs standard care, radiation cystitis	Improved urinary symptom score	Supports hyperbaric oxygen

9 Occupational Health in Urology

1 · Physical ergonomics

P1 Operating room set-up — monitor straight ahead with its top at eye level, elbows 90–120°, laparoscopic table at the surgeon's pubic symphysis

Not in the paper answer:

- Work-related musculoskeletal disorders affect **46–90%** of urologists. They are a cumulative-load problem, so every fix either restores neutral posture or interrupts the load.
- Posture targets: in open surgery, work **5–10 cm** below the elbow with neck flexion **<25°**. In endourology, set the target between symphysis and umbilicus. At the robotic console, keep back flexion **<15°** and knees at 90–120°.
- Use steep-declination loupes (neck **<20°**) and two-piece lead that puts weight on the hips. A presbyopic surgeon should use workspace lenses, not progressives.

2 · Occupational exposures

P1 P4 Operating room set-up; Operating room safety: the role of surgeon — as low as reasonably achievable (ALARA: time, distance, shielding), lead and dosimeter, smoke evacuation, sharps safety, vaccination, "sterile cockpit" for critical steps

Not in the paper answer:

- **Fluoroscopy**: use pulsed screening at the lowest rate (1 pulse/s), collimate, and use last-image hold. Keep the source far from the patient and the image intensifier close. **Magnification increases dose.**
- **Pregnant surgeon**: wear a fetal dosimeter at the waist under the apron, use maternity or double-thickness lead, and minimise radiation cases.
- For CO_2 laser on condylomata, wear an N95 mask with the evacuator tip within **5 cm**. For intravesical chemotherapy, wear two pairs of chemotherapy gloves and use a closed-system transfer device.

3 · Physical and mental health

- **Atherosclerotic cardiovascular disease** is the most threatening modifiable cause of death in surgeons. Recommended sleep is **7–9 h**, and exercise **150 min** moderate weekly plus strength training twice.
- **Burnout** means emotional exhaustion, depersonalisation and low personal accomplishment, and affects about **50%** of urologists. Spending **≥20%** of time on meaningful work lowers it.
- A **second victim** develops guilt and post-traumatic stress after an adverse event and needs psychological first aid and peer support.

Stress zone	Features	Response
Ready (green)	Balanced, focused	Exercise, sleep, mindfulness
Reacting (yellow)	Irritable, poor concentration	Self-care, delegate
Injured (orange)	Anxiety, substance misuse, burnout	Support, less workload, time off
Ill (red)	Panic, severe depression, dependence	Mental-health professional, sick leave

Imposter syndrome is a cognitive distortion, commonest in trainees, and responds to cognitive behavioural therapy.

10 Ethics and Informed Consent

1 · Principles of medical ethics

P1 Ethics and informed consent in urology — the four principles (Beauchamp and Childress, 1979), Hippocratic origins, Jonsen's four-box method

Not in the paper answer:

- All four principles are equal and **prima facie binding**; an ethical dilemma is a conflict between them that needs prioritising. Equal treatment is not always just treatment.
- In the four boxes, **quality of life draws on three principles** (autonomy, beneficence, non-maleficence) and **contextual features map to justice**.

2 · Informed consent

P1 As above — Schloendorff, Salgo ("informed consent") and Canterbury v Spence (reasonable-patient standard); four elements, with checking understanding most often missed; material risk ≈1%; decision aids, repeat-back and shared decision making

Not in the paper answer:

- The law moved from **battery to negligence to a patient-centred standard**. Mohr v Williams (1905) held that consulting a doctor is not consent.
- **Disclosing uncertainty improves trust.** Decision aids reduce decisional conflict; repeat-back works but is time-consuming.

3 · Capacity, exceptions and special situations

P1 As above — presumed consent in emergencies, refusal by a conscious capacitated patient, surrogate using substituted judgement then best interests, parental consent with child assent

Not in the paper answer:

- If a surrogate does not follow the patient's wishes, go to the **ethics committee** and then a court-appointed conservator.
- The **Patient Self-Determination Act (1990)** requires institutions to tell patients of their right to an advance directive.
- For research, the institutional review board may **waive consent** for minimal-risk or de-identified data.

11 Perioperative Care in Urologic Surgery

1 · Preoperative evaluation

- Test selectively: an electrocardiogram at age **>40** or with cardiac history, clotting tests only with a bleeding or anticoagulant history, and a pregnancy test in every woman of childbearing age.

American Society of Anesthesiologists (ASA)	Definition
I / II	Healthy / mild systemic disease
III	Severe disease that limits activity, not incapacitating
IV	Incapacitating, a constant threat to life
V / VI	Moribund (death within 24 h) / brain-dead donor
E	Emergency suffix (never with VI)

2 · Cardiac and other risk

- Good reserve with low-stress (endoscopic) surgery needs no further testing, whatever the history. Capacity of **≥4 METs** (metabolic equivalents, about two flights of stairs) means no invasive work-up.
- The **Revised Cardiac Risk Index** has six unweighted predictors: ischaemic heart disease, heart failure, cerebrovascular disease, high-risk surgery, preoperative insulin and creatinine **>2 mg/dL**.
- After myocardial infarction (MI), waiting 4–6 weeks before elective surgery is reasonable. Frailty predicts complications independent of age.
- A forced expiratory volume in 1 s **<0.8 L** or <30% predicted is high pulmonary risk. Child class mortality is A 10%, B 30% and C 76–82%.
- Loss of >9 kg in 3 months is severe malnutrition; preoperative parenteral nutrition helps only these patients.

P2 Stones (urolithiasis) in pregnancy — ultrasound first, fetal dose limit, no NSAIDs, second trimester safest

Not in the paper answer:

- The fetal malformation threshold is **10 cGy**; computed tomography gives about 1 cGy.

3 · Optimising comorbidity

- Only β-blockade has improved cardiac outcome (Mangano atenolol trial).
- Stop smoking **≥8 weeks** before surgery; stopping later may raise pulmonary risk.
- Target glucose 80–250 mg/dL. Withhold metformin on the day of surgery and sodium-glucose cotransporter-2 inhibitors 3 days before (euglycaemic ketoacidosis).
- Presume adrenal suppression with prednisone >20 mg/day for >3 weeks or cushingoid features; ≤5 mg/day needs only the usual dose. Give hydrocortisone **50–100 mg** intravenously at induction, then 25–50 mg 8-hourly for 24–48 h.

4 · Antithrombotic management

- Stop warfarin 5 days before (international normalised ratio **<1.5**). Bridge moderate- and high-risk patients with therapeutic low-molecular-weight heparin (LMWH), last dose 24 h before. Direct oral anticoagulants need no bridging.

Thrombotic risk	Mechanical valve	Atrial fibrillation ($CHADS_2$)	Venous thromboembolism (VTE)
High	Any mitral; caged-ball or tilting-disc aortic; stroke <6 months	5–6; stroke <3 months	<3 months; severe thrombophilia
Moderate	Bileaflet aortic plus risk factor	3–4	3–12 months; active cancer
Low	Bileaflet aortic alone	0–2	Single VTE >12 months

Drug	Stop before surgery
Aspirin, clopidogrel	7–10 days
Prasugrel / ticagrelor	7 / 5 days
Apixaban, rivaroxaban	48 h (72 h if high bleeding risk)
Dabigatran	48–96 h by creatinine clearance

> **KEY POINT**
> Stopping antiplatelets early after a coronary stent carries a 25–50% risk of MI. Delay elective surgery ≥30 days after a bare-metal stent and >1 year after a drug-eluting stent.

5 · Bowel preparation, temperature and positioning

P3 Methods of preoperative bowel preparation — mechanical versus oral antibiotic preparation, Nichols–Condon, enhanced recovery, alvimopan

P1 Patient positioning in lap urology; Operating room set-up — padding, abduction ≤90°, no shoulder braces; hypothermia triples wound infection

Not in the paper answers:

- Ileus beyond 72–96 h should prompt a search for obstruction or retroperitoneal haemorrhage.
- Femoral neuropathy comes from retractor blades on the psoas, not from hip position.

6 · Antibiotic prophylaxis

- Give it within **60 min** before incision, limit it to **≤24 h**, and redose after 4 h. Transrectal biopsy is a contaminated wound.

Procedure (American Urological Association, AUA)	First choice	Duration
Cystoscopy (sterile urine), shock-wave lithotripsy	Only with risk factors	Single
Endourology, nephrectomy, prostatectomy, ileal conduit	Cefazolin	Single
Transrectal biopsy	Fluoroquinolone, or cephalosporin plus aminoglycoside	Single
Colon conduit	Cefazolin plus metronidazole	Single
Percutaneous renal surgery	First- or second-generation cephalosporin	≤24 h

Procedure (American Urological Association, AUA)	First choice	Duration
Prosthesis	Aminoglycoside plus cephalosporin or vancomycin	≤24 h

7 · VTE prophylaxis

- VTE is the leading non-surgical cause of death after urological surgery, highest after cystectomy. Drug prophylaxis raises minor, not major, bleeding.
- Enoxaparin is **40 mg** daily (30 mg if creatinine clearance <30 mL/min). Unfractionated heparin (UFH) 5000 U is reversible and safe in renal failure.

AUA risk	Definition	Prophylaxis
Low	Minor surgery, <40 y	Early ambulation
Moderate	40–60 y, or minor surgery with risk factors	UFH 12-hourly or enoxaparin
High	>60 y, or 40–60 y with risk factors	UFH 8-hourly or enoxaparin
Highest	Cancer, prior VTE	Drug plus compression

A Caprini score **≥5** is highest risk. Give extended **28-day** LMWH after radical pelvic cancer surgery.

8 · Anaesthesia and transfusion

- Extubate only when the train-of-four ratio reaches **0.9**. Nitrous oxide expands closed gas spaces.
- Transfusion is often indicated at haematocrit **<21%** and rarely above 30%. Give platelets below 50,000 (100,000 for high-risk surgery).
- Transfusion-related acute lung injury is the commonest cause of transfusion death. Delirium may be the first sign of sepsis.

12 Fundamentals of Intraoperative Patient Safety

1 · Checklists, site verification and team safety

P4 Operating room safety: the role of surgeon — World Health Organization checklist (deaths down 47%, major complications down 36%), Universal Protocol, retained items, fire triangle, sharps, sterile cockpit, TeamSTEPPS, I'M SAFE

Not in the paper answer:

- Wrong-site surgery occurs in about **1 in 112,000** cases, and up to 30% of root causes arise before admission.
- Mark only the correct side, never with an "X". For lower-tract endourology, mark the wristband, not the meatus. Use a professional interpreter, never family.
- Simulation-based team training changes behaviour; didactic teaching alone does not.

2 · Retained items and fire

- A retained surgical item is one found after skin closure, even in theatre. It occurs in **1 in 2800–10,000** cases; sponges are commonest, then drains, then needles.
- A correct count does not exclude a retained item, so obtain a radiograph when in doubt.
- The **Silverstein fire-risk score** gives one point each for open oxygen, an ignition source and a site above the xiphoid; **3** means high risk. Monopolar diathermy causes about 90% of fires.
- For buccal graft harvest, keep inspired oxygen ≤30%, stop nitrous oxide and warn anaesthesia before using diathermy.
- Place the diathermy return pad in full contact over muscle, away from metal implants, and prefer bipolar with pacemakers.

ALGORITHM

1. Fire: stop the procedure and pour saline or water.
2. Remove the drapes, then use an extinguisher.
3. Follow RACE: Rescue, Alert, Confine (gases off), Extinguish or evacuate.

3 · Positioning injuries (see Ch 11)

- In lithotomy, stirrups compress the common peroneal nerve at the fibular head and cause foot drop. Hip flexion with knee extension stretches the sciatic nerve.
- **Well-leg compartment syndrome** follows prolonged lithotomy (**>4 h**) with raised legs and Trendelenburg. Prevent it by limiting time or lowering the legs periodically. Suspect it with pain out of proportion and a raised creatine kinase, and treat by urgent fasciotomy.
- In the flank position, long cases and obesity risk rhabdomyolysis.

13 Incisions, Exposure and Drains

1 · Abdominal incisions

Incision	Indications	Notes
Lower midline	Prostatectomy, cystectomy, diversion, pelvic node dissection	**Commonest** urological incision
Upper or full midline	Retroperitoneal lymph node dissection (RPLND), trauma, bilateral nephrectomy	**Most versatile**; poor in obesity
Pfannenstiel	Pelvic surgery, specimen extraction	Rectus separated, not cut; **least hernia**
Gibson	Renal transplant, lower ureter	Extraperitoneal; hockey-stick extension adds hernias

- Close midline fascia with strong slowly absorbable suture, taking bites **1 cm** from the edge and 1 cm apart, with suture length to wound length **≥4:1**.
- No technique is superior for dehiscence or hernia, but avoid continuous rapidly absorbable suture.

2 · Approaches to the kidney

Incision	Indications	Trade-off
Midline transperitoneal	Trauma, inferior vena cava (IVC) thrombus, bilateral disease, horseshoe	Early vascular control; poor upper retroperitoneum
Subcostal or chevron	Radical nephrectomy, bilateral tumours, IVC thrombus	Early vascular control; liver, spleen and bowel at risk
Thoracoabdominal (9th–10th rib)	Level III thrombus; large upper-pole or adrenal mass	Best exposure; pain no worse than flank
Flank (12th supracostal or 11th transcostal)	Partial or simple nephrectomy, xanthogranulomatous pyelonephritis	Extraperitoneal; **pleura** at risk; transcostal causes flank bulge
Transverse upper abdominal	Paediatric Wilms tumour	–

KEY POINT

An anterior incision gives bilateral reach and early pedicle control at the cost of the bowel; a flank incision stays extraperitoneal at the cost of the pleura.

ALGORITHM

1. Pleural tear: pass a red rubber catheter through the tear and tie a purse-string around it.
2. Put the catheter end under water and give sustained inflation.
3. Withdraw it as bubbling stops. A large tear needs a chest tube.

- A routine chest radiograph after nephrectomy is unnecessary.

3 · Inguinal and scrotal incisions

- Radical orchidectomy uses an inguinal incision of 5–7 cm. **A scrotal incision is contraindicated in suspected testis cancer.**
- Subinguinal inguinal lymph node dissection risks flap necrosis, infection and lymphocele.

4 · Drains

- A Penrose drain is passive and suits the scrotum. Jackson-Pratt and Blake drains are active closed suction and suit most other surgery.
- Use a separate stab incision away from the inferior epigastric vessels, and secure the drain with non-absorbable suture.
- Routine drains are questioned after robotic prostatectomy, partial nephrectomy and cystectomy; drain when the findings demand it.

14 Sutures and Suturing

1 · Choosing a suture

- Braided suture knots securely but harbours bacteria; monofilament inflames less but has more memory.

Suture	Strength retained	Absorbed by
Plain / chromic gut	7–10 d / 21–28 d	70 / 90 d
Caprosyn	~10 d	56 d
Monocryl (poliglecaprone 25)	2 wk 20–40%	91–119 d
Vicryl (polyglactin 910, braided)	3 wk 40–50%	56–70 d
PDS (polydioxanone)	6 wk 35–60%	182–238 d
Maxon (polyglyconate)	6 wk 25%	180 d
Prolene, Mersilene (braided polyester)	Indefinite	Never

KEY POINT

Strength is lost long before absorption. Match the support needed: skin 1–2 weeks (Monocryl, gut), soft tissue 2–4 weeks (Vicryl), fascia ≥6 weeks (PDS, Maxon).

- Chromic gut is the most inflammatory absorbable suture; Caprosyn absorbs fastest.
- Silk marks the cord at orchidectomy before a possible retroperitoneal lymph node dissection.
- **Never leave non-absorbable suture or exposed metal staples in contact with urine**, because they form a stone nidus.
- V-Loc barbed suture is not for fascia; Stratafix Symmetric PDS is the only barbed suture strong enough for fascia.

2 · Needles and knots

- A taper point is commonest. A cutting needle has its edge on the inner curve; a **reverse cutting** needle has it on the outer curve and suits retention sutures.
- The smallest taper needle is RB (17 mm). A UR or GU 5/8-circle needle suits the deep pelvis.
- Prolene knots unravel, so add extra throws.

3 · Techniques

- A vertical mattress everts the edges; a figure-of-eight gives haemostasis.
- Run fascia only with slowly absorbable suture, and close port defects >8 mm.

Trial	Design	Result	Take-home
STITCH	Randomised, midline: 5 mm versus 1 cm bites	Hernia at 1 year 13% vs 21%	Small bites

15 Lower Urinary Tract Catheterisation

1 · Urethral catheters

P1 Foley's (urethral) catheters, guidewires and DJ stents — calibre, French sizing, catheter types and materials, retrograde urethrogram first for suspected urethral injury, technique

Not in the paper answer:

- Lapides showed that infection comes from stasis and overdistension, not bacteria alone, so **regular emptying, not sterility, prevents infection** in clean intermittent catheterisation.

Age	Straight (Fr)	Balloon (Fr)
Newborn	4–6	6
1–23 months	6–8	6–8
2–6 y	10	8–10
7–12 y	10–12	12–14
>12 y	14	14–16

2 · Difficult catheterisation

P1 Outline algorithm for difficult catheter placement — blind, then wire, then vision, then suprapubic; 18 Fr coudé for prostatic enlargement; dilate and pass a Councill over the wire; special situations

Not in the paper answer:

- Membranous strictures are the most prone to false passage. Pass the cystoscope with the camera angled anteriorly.
- With a neobladder, suprapubic puncture is hazardous (intraperitoneal), so use image guidance.

3 · Suprapubic catheterisation

- It is indicated when the urethra is inaccessible (trauma, stricture) and for long-term drainage.
- Percutaneous placement is contraindicated with a violated retropubic space, bowel over the bladder, or no ultrasound.

ALGORITHM

1. Confirm a full bladder with no overlying bowel on ultrasound, and tilt the patient head-down.
2. Make a 1–2 cm incision 1–2 fingerbreadths above the symphysis.
3. Aspirate urine with a spinal needle while infiltrating local anaesthetic.
4. Insert a trocar and sheath (16 Fr catheter) or use the Seldinger technique, without over-advancing.

- Bowel injury is the key complication. An experienced person should make the **first change at 4–6 weeks**.

4 · Complications

P1 Catheter Associated Urinary Tract Infections (CAUTI) — 75% of hospital urinary infections; prevention bundle; treat only symptomatic infection

Not in the paper answer:

- Spinal cord injury raises bladder cancer risk, but surveillance cystoscopy is not recommended.

Retained catheter	Management
Balloon will not deflate	Cut the valve; pass a stiff wire end up the channel to puncture it, or use an ultrasound-guided needle; then cystoscopy to retrieve fragments
Knotted	Cystoscopic needle puncture of the balloon
Calcified	Cystoscopic laser lithotripsy

16 Fundamentals of Upper Urinary Tract Drainage

1 · Indications

- Obstruction is the commonest indication. Others are pre-stenting before retrograde intrarenal surgery, trauma (to prevent urinoma) and instillation of chemotherapy.
- In the **Whitaker test**, a renal pelvic minus bladder pressure **>12 cm H_2O** during constant-rate infusion means obstruction.

P1 Genitourinary infections, bacteraemia and urosepsis; Interventional radiology in urology — the obstructed infected kidney needs urgent drainage; retrograde stent equals percutaneous nephrostomy (PCN); no definitive stone surgery during sepsis

Not in the paper answer:

- PCN avoids general anaesthesia in the very ill, while a stent avoids an external tube and can also be placed awake under local anaesthesia.

2 · Anatomy chooses the route

Situation	Preferred approach
Horseshoe kidney	Percutaneous via a more medial upper-pole calyx; computed tomography (CT) first, as retrorenal bowel is common
Pelvic or ectopic kidney	Supine anterior, image-guided or laparoscopy-assisted
Transplant	Anterior, ultrasound-guided
Ureteropelvic junction obstruction before pyeloplasty	Antegrade, to spare the ureter
Calyceal diverticulum, ileal ureter, transureteroureterostomy, urinary diversion	Antegrade

3 · Retrograde stenting and stents

P1 P3 Foley's catheters, guidewires and DJ stents; Discuss various ureteric stents — materials, metallic stent for extrinsic compression, stent symptoms, complications by timing, encrustation and FECal grade, forgotten stent, change every 3–6 months

Not in the paper answer:

- Adult stents are typically 4.7–8 Fr and 22–30 cm; **6 Fr, 24–26 cm** is commonest.
- Under fluoroscopy alone, place the pusher marker at the mid-symphysis in men and the lower symphysis in women. This landmark is invalid after prostatectomy or with prolapse.
- The Ureteral Stent Symptom Questionnaire (USSQ) has **38 questions in six domains**. Stent length made no difference in a large randomised trial.

4 · Percutaneous nephrostomy

P1 Pelvicalyceal anatomy and its relevance to PCNL; Prone vs supine PCNL — posterior calyx through the papilla along Brödel's avascular line, never the pelvis; calyx choice and pleural risk; prone versus supine

Not in the paper answer:

- The American Urological Association (AUA) recommends antibiotic prophylaxis for all percutaneous renal surgery.

- A 21 G needle takes only a **0.018-inch** wire, which must be upsized; an 18 G needle takes a 0.035-inch wire. Dilate short, never beyond the target calyx.
- Balloon dilators are now commonest and bleed less but work poorly in scar, where metal Alken dilators are best.

Agent	Stop before access
Aspirin (primary prevention only)	1 week; continue if there is a cardiovascular indication
Warfarin	1 week (bridge if needed)
Clopidogrel	5 days
Non-steroidal anti-inflammatory drugs	3–5 days

Tube	Size	Notes
Foley or Councill	≤24 Fr	Fill balloon partly with saline, not contrast
Cope loop	6–14 Fr	Dislodges in obesity
Nephroureteral	8–12 Fr	Rarely dislodges
Malecot	16–30 Fr	Needs a stiffener

- Smaller tubes cause less pain and leak, with no more bleeding. Clamp before removal to test distal patency, and **exchange every 3 months**.

P2 Tubeless PCNL — benefit is over large tubes, not 8–9 Fr tubes; keep a tube for pyonephrosis, bleeding or perforation

5 · Complications

P2 Complications of PCNL — immediate venous bleeding needs tamponade, bleeding at about 1 week is a pseudoaneurysm or arteriovenous fistula needing selective embolisation; pleural, colon and solid-organ injury

Not in the paper answer:

- Major complications occur in about 10%; PCN transfusion is **0.5–4%**. Pleural injury needing intervention is about **5%** above the 12th rib and **25%** above the 11th.
- Colon injury occurs in about 0.1%; retrorenal colon is found in 10% prone versus 2% supine. Extraperitoneal injury is managed conservatively; peritonitis needs open repair.
- Air embolism shows as a sudden fall in end-tidal CO_2 and a mill-wheel murmur. Give 100% oxygen and position head down, right side up.

17 Principles of Urologic Endoscopy

1 · Evolution of the endoscope

P4 P1 Flexible endourological instruments, digital and current ureteroscopes; Pioneers of urology — Tyndall, Marshall (1964), Hopkins rod-lens (1966), digital and single-use scopes

Not in the paper answer:

- **Desormeaux (1853)** excised a urethral papilloma, the **first therapeutic endoscopy**. Nitze (1877) placed the light source at the tip.

2 · Optics and enhanced imaging

P4 Flexible endourological instruments — coherent versus incoherent fibres, chip-on-tip sensors, fibreoptic versus digital

P2 CA bladder, etiology, newer investigations and treatment — blue-light cystoscopy with hexaminolevulinate, and narrow-band imaging at 415 and 540 nm

Not in the paper answer:

- Narrow-band imaging (NBI) needs **no instillation**.
- Blue-light cystoscopy (Food and Drug Administration approved 2010): hexaminolevulinate is instilled **1 h** before, and blue light of 360–450 nm makes tumour fluoresce red while normal urothelium stays blue.

3 · Irrigation fluids and TUR syndrome

P2 P1 Irrigation fluids for TURP; TUR Syndrome — fluids and osmolality, mechanism, risk factors, prevention, 3% saline

Not in the paper answer:

- **Ureteroscopy always uses normal saline**, so that no hypotonic fluid is absorbed. Saline's only hazard is volume overload, without hyponatraemia.
- Serum sodium is usually **<125 mmol/L** in transurethral resection (TUR) syndrome. This chapter advises a correction of no more than 8–10 mmol/L in 24 h (the paper answer gives ≤12).

4 · Cystourethroscopy

P1 Indications for outpatient cystourethroscopy — diagnostic and therapeutic indications, lidocaine dwell ≥5 min, no routine prophylaxis, no surveillance cystoscopy for chronic catheters (AUA/SUFU 2021)

P4 Flexible endourological instruments — flexible cystoscope 14–17 Fr, deflection 120–220°, channel 6.4–7.5 Fr; fibreoptic illuminates better than digital

Rigid instruments

- French (Charrière) gauge: **3 Fr = 1 mm** diameter. The diagnostic rigid sheath is 15–17 Fr. Choose the smallest scope that does the job.
- A rigid set has a lens, bridge, sheath and obturator. Use a 0° or 12° lens for the urethra, 25° or 30° for general work, and 70° or 120° for the dome, anterior wall and bladder neck.
- The **Albarran bridge** has a lever that deflects catheters and wires into an orifice behind a high bladder neck, a large median lobe or a reimplanted ureter.

Antibiotic prophylaxis: AUA versus EAU

Prophylaxis is driven by patient risk, not by the procedure.

	American Urological Association (AUA)	European Association of Urology (EAU)
Diagnostic cystoscopy	None unless patient risk factors are present	
TURP	Yes (<24 h)	Yes
TURBT	Yes	**Only if sepsis risk is high**
Ureteroscopy	**All patients**	According to local pathogens
Agents	Cefazolin or trimethoprim-sulfamethoxazole (ureteroscopy: TMP-SMX or a 1st/2nd-generation cephalosporin)	Local antibiogram
Added risk factors	Smoking, steroids, distant infection, prolonged stay, anatomical anomaly	Diabetes, weight extremes, prior urinary infection, previous instrumentation

Both share older age, malnutrition, immunodeficiency and catheters or colonised material. Treat any urinary tract infection before instrumentation.

Technique and special situations

- Prepare the genitalia with **aqueous povidone-iodine** at any age, because chlorhexidine and alcohol damage mucosa.
- **Orthotopic neobladder**: do not dilate or incise an anastomotic contracture without obstruction, as this risks incontinence. **Cutaneous pouch**: use a flexible scope through the channel for diagnosis, and treat percutaneously after imaging.

5 · Upper-tract endoscopy: indications

P2 ESWL: mechanism, indications, factors affecting outcome — AUA/EAU stone size thresholds by site, unfavourable stone and anatomical factors

P4 Flexible endourological instruments (role) — UTUC biopsy predicts grade in about 9 of 10, endopyelotomy, endoureterotomy for strictures <1.5 cm, essential haematuria

Not in the paper answer:

- The EAU prefers ureteroscopy (URS) for ureteric stones **>10 mm**.
- Diagnostic URS for upper tract urothelial carcinoma does not worsen survival but **raises intravesical recurrence**, so survey the bladder.
- Endoureterotomy also fails more with an allograft, an irradiated or ureteroenteric stricture, or a poorly functioning kidney. Balloon dilation alone is about 50% durable at 1 year.

6 · Ureteroscopes

P4 P2 Flexible endourological instruments; RIRS / flexible ureteroscopy — semirigid and flexible specifications, 3.6 Fr channel, ≥270° deflection, digital versus fibreoptic, single-use scopes, laser care

Not in the paper answer:

- **Passive (secondary) deflection** off the upper pelvic wall lengthens the effective tip, but it **fails in gross hydronephrosis**, where there is no wall to push against.

7 · Accessories

P1 Foley's (urethral) catheters, guidewires and DJ stents — wire types, sizes and uses

P4 P3 LASERs in urology; Therapeutic energy sources — holmium 2100 nm, thulium fibre laser 1940 nm, pneumatic and electrohydraulic lithotripsy, retropulsion

Not in the paper answer:

- The need for dilation falls with scope size (**<1%** for a 7.4 Fr scope). A **zero-tip balloon** dilates right up to an impacted stone.
- **Electrohydraulic lithotripsy is not first line in the ureter** (AUA), because its poorly focused shock damages the ureter and scope. A rigid pneumatic probe needs a rigid scope.
- The **three-prong grasper is the safest retrieval device**, because it releases an oversized stone. Never basket an irretrievably large stone through a single-channel flexible scope.

8 · Ureteroscopy technique

P2 RIRS / flexible ureteroscopy — sterile urine, safety wire, access sheath, laser power limits, relocation of lower-pole stones, when to omit a stent

ALGORITHM

1. Perform cystoscopy and place a 0.038-inch PTFE safety wire.
2. For an impacted stone, pass an angled hydrophilic wire (with or without a torque catheter), then exchange it for a PTFE wire.
3. If obstruction with infection is suspected, aspirate the pelvis. If the aspirate is turbid, place a stent and postpone.
4. Drain the bladder and proceed: semirigid scope below the iliac vessels, flexible above.

- If the orifice will not admit the scope, the safest course is to **stent and return in 1–2 weeks**. **Never dilate a segment that contains a stone**, because it may perforate.
- Formal dilation is needed in about 8–25%. Fragment an impacted stone to dust of **≤2 mm**, or basket the fragments.
- Under-rotation is the commonest novice error, and over-rotation damages fibreoptic bundles. Use a 200 µm fibre when deflection is needed.
- The **T-tilt** (15° Trendelenburg plus 15° airplane away) helps at the lower pole. Give non-steroidal anti-inflammatory drugs, with or without an alpha-blocker, after surgery.

9 · Nephroscopy and complications

- The rigid nephroscope (about 24–26 Fr) passes through a 26–30 Fr Amplatz sheath, and a flexible nephroscope reaches calyces out of line.

P2 RIRS / flexible ureteroscopy — sepsis about 5%, stricture about 3%, perforation, avulsion

Not in the paper answer:

- For mucosal injury or perforation, stop and stent for 2–4 weeks. Avulsion follows dragging an oversized basketed stone, and needs open or robotic reconstruction.

18 Fundamentals of Laparoscopic and Robotic Urologic Surgery

1 · Patient selection and contraindications

Absolute contraindications	Relative contraindications (caution, specific consent)
Uncorrectable coagulopathy; intestinal obstruction (unless being treated); abdominal wall infection; massive haemoperitoneum or haemoretroperitoneum; generalised peritonitis; suspected malignant ascites	Morbid obesity; extensive previous surgery; pelvic fibrosis; organomegaly; benign ascites; pregnancy; hernia; iliac or aortic aneurysm; surgeon inexperience

KEY POINT

Every relative contraindication is a problem of safe entry or of tolerating pneumoperitoneum. The answers recur: enter elsewhere (Palmer point, retroperitoneum), enter openly (Hasson), or insufflate less.

- Severe chronic obstructive pulmonary disease (COPD) needs arterial blood gases and lung function tests. Treat arrhythmias first, because hypercarbia and acidosis destabilise the myocardium.
- **Pregnancy**: enter cephalad to the fundus (Palmer point), use **8–12 mm Hg**, and operate in the **second trimester**; it is harder after 20 weeks.
- A diaphragmatic hernia lets carbon dioxide (CO_2) reach the mediastinum. Never insufflate through an abdominal-wall hernia.
- In morbid obesity, renal and adrenal laparoscopy still beats open surgery. Leave prostatectomy and cystectomy with body mass index (BMI) >30 until later in the learning curve.

2 · Preoperative preparation and set-up

P1 Operating room set-up; Patient positioning in lap urology — pre-insufflation checklist, robotic cart layout, positions and their hazards, axillary roll, no shoulder braces, rhabdomyolysis risk

Not in the paper answer:

- **No mechanical bowel preparation** is needed unless bowel is used, and it shows no benefit even for cystectomy.
- Enhanced recovery after surgery (ERAS) allows solids until 6 h and a carbohydrate drink 2 h before surgery. It shortens stay by about **30%** and halves complications.
- Group and screen for nephrectomy, nephroureterectomy and prostatectomy. Crossmatch for partial nephrectomy, cystectomy and caval thrombectomy early in experience.

3 · Physiology of pneumoperitoneum

P1 Pneumoperitoneum physiology, complications of retroperitoneoscopy and laparoscopic donor nephrectomy — two drivers, insufflants, pressure table, effects by system and position

Not in the paper answer:

- Helium or argon can be switched to **to salvage intraoperative hypercarbia**.
- The commonest working pressure is **15 mm Hg**. A pressure of ≤10 mm Hg markedly reduces oliguria.

4 · Access

P1 Laparoscopic access (Veress, Hasson cannula, extraperitoneal space), gases — Veress sites and tests, Hasson steps, balloon dissection, balloon-tip port

Not in the paper answer:

- The Veress needle should give flow with ≤2 mm Hg at 2 L/min. **Do not lift the fascia**, because this widens the gap between fascia and peritoneum rather than the abdominal space.
- At the umbilicus, angle towards the pelvis in thin patients and go perpendicular in obese patients, as the umbilicus migrates caudally.
- A **hand port placed first** is the safest entry for the inexperienced, as the intra-abdominal hand guides the secondary trocars.

	Transperitoneal	Retroperitoneal
Space	Large, familiar anatomy	Small; psoas is the key landmark; direct (artery-first) hilar access
Physiology	Larger haemodynamic swings	More emphysema and pneumothorax; CO_2 absorption similar with a sealed port
Best for	Large tumours, reconstruction	Previous transperitoneal surgery; posterior tumours

Vascular injury in upper-tract surgery is about **5×** the pelvic rate.

5 · Trocars and port placement

P4 Different types of trocars in laparoscopy — bladed versus blunt (with figures), optical, balloon and AirSeal ports, capacitive coupling

P4 Laparoendoscopic single-site surgery: LESS — clashing, articulating instruments, da Vinci SP

Not in the paper answer:

- **Triangulation**: place ports in a diamond around the target, with instruments at **<55°** to the horizontal and 25–45° between them. Place ports in the midline or at least 6 cm lateral.
- Before secondary trocars, inspect the whole abdomen for entry injury. Aim each port at the target, as redirecting widens the tract and causes emphysema.
- **Hand port**: the incision in centimetres equals the surgeon's glove size. Wound infection is about 9% and midline hernia 4–7%.

6 · Robotic systems and instruments

P4 P3 Robotics in urology; Hemostasis in lap surgery; Therapeutic energy sources — da Vinci systems, EndoWrist, robot failure; energy-device lateral spread; staples, clips and Hem-o-lok rules

Not in the paper answer:

- EndoWrist instruments have about a **10-use** limit, and an Allen bolt opens jaws locked after failure.
- Robot camera ports are 8 mm (Xi) or 12 mm (Si). Place the assistant port caudal, not between the arms.
- The LapSac is the most tear-resistant bag for morcellation, and ring forceps puncture it least.

7 · Exiting the abdomen

ALGORITHM

1. Lower the pressure to 5 mm Hg to unmask bleeding.
2. Inspect the operative site and every port.
3. Place fascial sutures under vision and remove each port under vision, the scope last.
4. Close a hand port first, then reinsufflate.

Trocar	Adult	Child
Bladed	Close all 10–12 mm sites	Close all ≥5 mm sites
Blunt	Close midline >10 mm or stretched sites	

- With BMI >30, use a closure device. Transverse extraction incisions herniate less than vertical midline ones.

8 · Complications and management

P4 P1 General complications related to laparoscopy; Pneumoperitoneum physiology and complications of retroperitoneoscopy — complications by phase, gas embolism, vessel injury, electrosurgical mechanisms, late thermal injury, venous bleeding, nerve injury, rhabdomyolysis

Not in the paper answer:

- **Bowel** is the commonest first-trocar injury. A through-and-through injury is seen only by viewing the primary port from a secondary port, so do this routinely. Leave the trocar in, gain access elsewhere and repair, with 4–5 L antibiotic irrigation. The commonest major vessels injured are the **aorta and common iliac artery**.
- **Thermal bowel injury**: resect a monopolar burn with **6 cm** margins, excise the visible area of a bipolar burn, and repair a mechanical injury laparoscopically.
- A Veress needle in a vessel is withdrawn without manipulation and re-sited. Always repair a bladder injury, and catheterise every major case.

9 · Training

P4 Simulators in urology — box trainers, virtual reality, animal and cadaver models, E-BLUS, Fundamentals of Robotic Surgery

Not in the paper answer:

- Robotic surgeons with **>20 cases** have better outcomes, so mentor early cases.

19 Energy Modalities in Urology

P3 P4 Therapeutic energy sources in urology; Basic energy modalities in urological surgery — classification (electrical, electromagnetic, acoustic, direct contact), electrosurgery, lasers, acoustic and ablative energies with present status

1 · Physics essentials

Every modality depends on where the energy is absorbed.

- The five surgical effects are cutting, ablation, destruction, coagulation and fragmentation.
- Laser power equals pulse energy times frequency, so 0.5 J × 20 Hz gives 10 W. **10 bar = 1 MPa** (not 1 bar).
- High frequency penetrates shallowly and low frequency deeply, which is why a scrotal probe is linear and a renal probe curvilinear. Absorption usually rises with frequency.
- Photon energy is E = hf, so raising laser power sends more photons per second, not more energetic ones.
- Acoustic impedance is sound speed times density. A large mismatch (tissue–air, tissue–stone) reflects the wave, which explains gel coupling, gas and rib shadows and the stone acoustic shadow.
- Radiation force equals absorbed power divided by wave speed. It is negligible for light but useful for sound, and is the basis of ultrasonic stone propulsion.
- A shock is an abrupt pressure rise from nonlinear steepening, and is acoustic, not electrical. Cavitation is a vapour bubble torn open by negative pressure, whereas boiling is caused by a rise in temperature.

	Electromagnetic waves	Elastic (acoustic) waves
Devices	Lasers, electrosurgery, microwave, radiofrequency (RF)	Shock wave lithotripsy (SWL), ultrasound therapies, intracorporeal lithotripters
Medium and wavelength	Free space; microns (light)	Needs gas, liquid or solid; millimetres
Bioeffect	Absorption, giving heat	Heat, mechanical stress in solids, cavitation in fluids
Penetration	Short, so a fibre must reach the target	Centimetres, so transcutaneous delivery is possible

2 · Electrosurgery

P2 P3 P4 Energy sources in laparoscopic urology; Therapeutic energy sources; Basic energy modalities — 500 kHz, current density, cut/coagulation/fulguration/blend waveforms and wattages, monopolar vs bipolar, vessel sealers, ultrasonic shears, lateral spread, the four mechanisms of stray-current injury

Not in the paper answer:

- **Pad placement**: use clean, well-perfused muscle and avoid bony prominences, scars, dense hair and metal implants. A **hip prosthesis is not a contraindication**; place the pad on the opposite side.
- **Pacemakers and implantable defibrillators**: prefer bipolar. If monopolar is needed, route the current path away from the device and use short bursts. Cardiology sets pacing to asynchronous and suspends anti-tachycardia therapy, an external defibrillator must be available, and the device is rechecked after surgery.

- **Surgical fire** needs fuel (drapes, alcohol prep), an oxidiser (oxygen-rich gas) and ignition. Electrosurgery is implicated in most theatre fires.

Endoscopic resection

P1 P2 P3 Monopolar v/s Bipolar TURP; TUR syndrome — ion-free 1.5% glycine vs saline, hyponatraemia, comparable efficacy, obturator reflex and its prevention

Not in the paper answer:

- Typical monopolar resection settings are cut **180 W** and coagulation **80–120 W**.
- Ions in the irrigant disperse current away from the loop, which is why monopolar irrigant must be ion-free, inert, non-toxic and transparent.
- Damaged sheath insulation causes loss of cutting or a urethral burn, so inspect the loop before mounting.

Device-assisted intravesical therapy for non-muscle-invasive bladder cancer (NMIBC)

	Electromotive drug administration (EMDA)	Chemohyperthermia
Energy	Electric field between an intravesical electrode and abdominal pads	915-MHz microwave via catheter (Synergo) with thermocouples
Mechanism	Iontophoresis, electro-osmosis and electroporation drive mitomycin C (MMC) into the lamina propria	Bladder wall at **42 °C** raises permeability and cancer-cell heat sensitivity
Evidence	Randomised controlled trial (RCT) of EMDA-MMC with bacille Calmette–Guérin (BCG): better recurrence- and progression-free survival. Meta-analysis: recurrence only. Cochrane: low quality	Not standard

3 · Lasers

P4 LASERs in urology and newer laser technologies; Holmium laser; Thulium laser; KTP laser — principle, wavelengths and penetration depths, photothermal vs photomechanical, Moses effect and pulse modulation, retropulsion, dusting vs fragmenting settings, heat (≤10 W in the ureter), thulium fibre laser vs holmium randomised trials, laser safety

P2 P3 P4 Lasers in BPH, including HoLEP; Green light lasers — enucleation and photovaporisation of benign prostatic hyperplasia (BPH)

Laser lithotripsy: optical pulse duration decides the mechanism

Optical pulse	Mechanism	Bubble pressure	Fragments
Ultrashort <500 ns (Q-switched)	Plasma	Extremely high	Large
Short 0.5–10 µs (pulsed dye; frequency-doubled double-pulse neodymium:YAG [Nd:YAG], FREDDY)	Cavitation and shock (photomechanical)	>300 bar	Large
Long >250 µs (holmium:YAG [Ho:YAG], thulium fibre laser [TFL])	Photothermal	<20 bar	Small, dust
>20 ms or continuous wave	Photothermal with carbonisation	–	Adjacent tissue injured

Not in the paper answer:

- Manufacturers quote the electrical (pump) pulse duration, but the **optical pulse duration** is the one that matters. Dosimetry, not the platform, sets the mechanism.
- **Moses effect**: the first ≈ **60 µs** boil a vapour channel. The fibre should be close and perpendicular to the stone (0° gives the most ablation; parallel gives none). Ho:YAG transients are only 8–20 bar at <1 J.
- The **pulsed dye laser (504 nm)** was safe because haemoglobin acts as an energy sink, but it failed on hard stones (cystine, brushite, calcium oxalate monohydrate).

Soft-tissue lasers

- The **CO_2 laser** (10,600 nm) is used for superficial penile lesions and cannot be used intracorporeally.
- The **diode laser** (940–1490 nm) is absorbed by water and haemoglobin variably, penetrates 0.5–5 mm and is used for prostate vaporisation.
- Nd:YAG forward scatter can injure bowel beyond the bladder wall. KTP is shallow because of haemoglobin absorption, not wavelength itself.
- **GreenLight** ablates best at a **4 mm/s** sweep and **≤2 mm** working distance. Gain plateaus beyond 180 W, but 180 W shortens operating time compared with 120 W.
- **Photodynamic therapy** treats the whole urothelium after one intravesical application of photosensitiser, by generating singlet oxygen.

4 · Acoustic therapy

Heating	Cavitation
Rises with frequency (absorption)	Rises as frequency falls (longer negative phase)
Dose-dependent	**Threshold effect**; nucleates at interfaces, stones and gas, so lung and bowel are vulnerable

- Do not perform SWL within **24 h** of ultrasound microbubble contrast. Doppler twinkling artefact is caused by microbubbles on the stone.
- Acoustic waves are non-ionising, so repeated safe exposures do not accumulate.

Shock wave lithotripsy

P2 P4 ESWL: mechanism, indications, factors affecting outcome, complications; Energy sources for breaking stones — electrohydraulic, electromagnetic and piezoelectric generators, F1 and F2, spallation and cavitation, 60–90/min better than 120/min, poor results with stones >1000 Hounsfield units; intracorporeal pneumatic, ultrasonic, combined and electrohydraulic lithotripters

Not in the paper answer:

- Pulse parameters: peak negative pressure ≈ **10 MPa** lasting ≈ 5 µs, about 3 J per pulse, and about **2000 shocks** at 0.5–3 Hz.
- Each shock leaves bubbles that need ≈ **1 s** to dissolve, so faster firing builds a cavitation cloud that shields the stone and injures the kidney. Small air pockets in the coupling disproportionately reduce focal pressure.
- Piezoelectric sources focus at the centre of curvature of a spherical bowl and are the only type that produces both impulses and narrowband bursts.

Investigational acoustic stone therapies

	Burst wave lithotripsy (BWL)	Ultrasonic propulsion
Waveform	Narrowband, 5–100 cycles, 200–1000 kHz	About 1500 cycles in bursts of ≤3 s
Peak pressure and rate	5–8 MPa; ≥10 Hz	≤2.4 MPa; 50% duty cycle
Guidance and anaesthesia	Real-time ultrasound; none	Ultrasound; none
Action	Breaks stones into uniform fragments proportional to wavelength	Moves fragments by radiation force

KEY POINT

Acoustic therapy is management of the negative-pressure phase. Tension cracks brittle stones and spares ductile tissue, but the same tension nucleates bubbles that either destroy tissue deliberately (histotripsy) or shield the stone and injure the kidney (SWL fired too fast). Raise the frequency for heat and lower it for cavitation.

Soft-tissue ablation

P2 HIFU — high-intensity focused ultrasound for prostate cancer and BPH

Not in the paper answer:

- **HIFU** uses 1500–10,000 W/cm² in ~1-s pulses and is poorly visible on ultrasound, so gaps occur.
- **Cavitation histotripsy** uses >10,000 W/cm² to make a bubble cloud that liquefies tissue mechanically, and is **echogenic in real time**. **Boiling histotripsy** boils the focus within one burst (<100 ms). Both are investigational.

5 · Thermal and other ablation

P2 P3 P4 Therapeutic energy sources; Basic energy modalities — radiofrequency ablation (60 °C, char at 150 °C), microwave ablation (less heat sink), cryoablation (<−40 °C, two freeze–thaw cycles), irreversible electroporation, Rezum, Aquablation; Biological effects of cryosurgery

Not in the paper answer:

- In radiofrequency ablation the tissue is heated by current, so the probe itself is not hot. Early concerns about microwave ablation (collecting-system absorption, inconsistent zones) were not borne out, and its safety and efficacy equal radiofrequency ablation.
- **Rezum** injects 103 °C steam through an 18-G needle; each **9-s** injection brings tissue to ≥70 °C and stays within the zonal compartment.

KEY POINT

Current density explains electrosurgery, pulse duration explains laser lithotripsy, penetration depth (chromophore) explains soft-tissue lasers, and negative pressure (frequency and rate) explains SWL and histotripsy. In direct-contact modalities the tissue boundary, not absorption depth, limits the lesion.

6 · Landmark trials

Trial	Design	Result	Take-home
TFL vs low-power Ho:YAG	RCT, ureteroscopy for renal stones	TFL: higher stone-free rate, shorter time, less bleeding	TFL beats 30-W holmium
TFL vs high-power Ho:YAG	RCT vs 120-W pulse-modulated Ho:YAG	Stone-free equal; TFL less retropulsion; Ho:YAG less energy	The comparator explains discordant trials
Ultrasonic propulsion of residual fragments	RCT	Fewer relapses (growth, urgent visit, surgery) at 4 years	Clearing residual fragments matters

20 Complications of Urologic Surgery

1 · Classification and context

P1 P4 Complications of retroperitoneoscopy; General complications related to laparoscopy — risk factors, complications by phase, vascular injury, gas embolism, thermal and missed bowel injury

- **Clavien–Dindo** is the recommended system and grades the **treatment required, not the event**. It omits intraoperative events and does not separate early from late.

Grade	Definition	Examples and traps
I	Deviation without drugs or intervention	Antiemetics, analgesics, electrolytes; wound opened at bedside
II	Other drugs	Blood transfusion, total parenteral nutrition
IIIa / IIIb	Surgical, endoscopic or radiological intervention	IIIa without, IIIb under general anaesthesia
IVa / IVb	Life-threatening, intensive care	IVa single organ (dialysis); IVb multiorgan. Central nervous system bleed or stroke counts; transient ischaemic attack does not
V	Death	–
Suffix "d"	Present at discharge	Flags follow-up; not a grade

2 · Positional and neuromuscular complications

P1 Patient positioning in lap urology — positions, nerve injury, rhabdomyolysis

Not in the paper answer:

- Positional injury occurs in ≈ **1%** of laparoscopic and up to ≈ **6%** of robotic cases.
- In lithotomy the stirrup compresses the **common peroneal nerve at the fibular neck** (foot drop). Tucked arms in robotic surgery endanger the **ulnar nerve**.

Compartment syndrome and operative nerve injury

- **Lower-limb compartment syndrome** occurs in 1 in 3500 overall and up to **1 in 500** radical cystectomies. Perform fasciotomy if >**30 mmHg** and symptomatic. Pressure >50 mmHg for 3–4 h causes irreversible necrosis.
- **Genitofemoral** injury (psoas hitch, ≈ 5%) is avoided by placing sutures along the tendon. Repair an **obturator** transection with tension-free **7-0** non-absorbable epineural sutures.

3 · Urine leaks and urinary tract injury

Urine leak after partial nephrectomy

It presents at about day 13. Drain creatinine confirms it but does not locate it; computed tomography (CT) urography with delayed images shows the site.

ALGORITHM

1. Drainage and a Foley catheter.
2. Ureteric stent; drain off suction.
3. Radiological drainage of collections, then nephrostomy.
4. Nephrectomy as the last resort.

Ureteric injury

P3 Ureteric injury: iatrogenic and upper ureteric; Uretero-vaginal fistula — danger sites, diagnosis, double-dye test, timing and repair by level

Not in the paper answer:

- A partial tear is closed with 4-0/5-0 absorbable sutures over a stent for 4–8 weeks.
- Strictures are treated endoscopically if short, and surgically if **>2 cm** or after endoscopic failure.

Bladder injury and anastomotic leak

- Close a bladder injury in two layers and catheterise for 7–10 days.
- A **urethrovesical leak** after robotic prostatectomy occurs in up to ≈ 15% at days 7–10.

> **ALGORITHM**
>
> 1. Keep the catheter patent and **never apply traction**.
> 2. Move the drain off suction; radiological drainage; then nephrostomy.
> 3. Formal repair of a major disruption **before day 7**, after which tissue will not hold sutures.

4 · Bleeding and vascular injury

P1 P3 P4 Retroperitoneoscopy and laparoscopy complications; partial nephrectomy — leave the trocar in place, venous bleeding control; delayed haematuria means pseudoaneurysm, so angioembolise

Not in the paper answer:

- Haemorrhage is the commonest intraoperative complication (≈ 2%).
- The commonest vessel injured at umbilical access is the **left common iliac vein**. Only the **internal iliac and inferior mesenteric** vessels may be ligated; all other major vessels are repaired.
- The superior mesenteric artery can be mistaken for the left renal artery in the lateral position, with mortality up to **75%**.

5 · Bowel injury

Most delayed injuries are thermal (monopolar spread >20 mm).

Injury	Management
Small tear without spillage	Primary repair
Colonic with faecal soiling	Colostomy
Thermal burn	**Always oversew, never conservative**; resect if extensive, because damage exceeds what is visible
Rectal (prostatectomy ≈ 0.5–1%; salvage ≈ 7%)	Multilayer closure, omental interposition; colostomy if irradiated or missed

Duodenal injuries perforate late even when <5 mm.

6 · Venous thromboembolism (VTE) and lymphocele

Symptomatic VTE occurs in ≈ **1%** after nephrectomy or prostatectomy and ≈ **5%** after radical cystectomy, mostly after discharge.

Caprini score: 1 point for age 41–60 or body mass index >25; 2 points for age 61–74, malignancy, or major or laparoscopic surgery >45 min; 3 points for age ≥75, personal or family

VTE or thrombophilia; 5 points for stroke, arthroplasty or fracture. Risk is very low at 0, low at 1–2, moderate at 3–4 and **high at ≥5**.

Risk	Prophylaxis
Very low (transurethral, anti-incontinence)	Early ambulation
Low	Intermittent pneumatic compression (IPC)
Moderate	Low-molecular-weight heparin (LMWH), low-dose unfractionated heparin or IPC
High	Heparin plus IPC; IPC alone until haemostasis is secure if bleeding risk is high
High-risk cancer surgery	Add extended LMWH for ≈ **4 weeks** after discharge

Start pharmacological prophylaxis 2–12 h before surgery.

P1 Lymphocele and postoperative lymphatic leak — incidence, observation, drainage, fenestration

Not in the paper answer:

- Prevent it by clipping or sealing visible lymphatics and by a peritoneal flap after robotic prostatectomy.

7 · Landmark trials

Trial	Design	Result	Take-home
Preventer	Randomised, perioperative heparin after prostatectomy	No reduction in symptomatic VTE	Not mandatory in low-risk prostatectomy
Extended LMWH meta-analysis	Randomised trials, abdominopelvic cancer surgery	≥1 month of LMWH cut VTE without extra bleeding	Extend after cystectomy

21 Urologic Considerations in Pregnancy

1 · Physiological changes

System	Change	Urological relevance
Cardiovascular	Cardiac output up 30–50%	The uterus compresses the vena cava, so use a **15–20° left tilt** for every procedure
Haematological	Blood volume up 40–50%; hypercoagulable	Thromboembolism risk ~10×; white cell count up to 12,200, so it is unreliable for infection
Renal	Glomerular filtration rate up **40–65%**; creatinine ≈ **0.8 mg/dL**	A "normal" non-pregnant creatinine may mean impairment
Urine	Protein up to 300 mg/24 h; calciuria doubles	Persistent proteinuria: exclude pre-eclampsia, diabetes, renal disease and infection

Dipstick microscopic haematuria occurs in up to **20%** and is usually benign; pregnancy-specific causes are pre-eclampsia and placenta percreta.

2 · Physiological hydronephrosis

P1 P2 Bacteriuria in pregnancy; Stones (urolithiasis) in pregnancy — mechanism, right-sided dilatation at the pelvic brim, Doppler resistive index to separate it from obstruction

Not in the paper answer:

- It peaks in the second trimester and resolves by up to 12 weeks postpartum; its grade correlates poorly with pain.
- Symptomatic hydronephrosis without a stone is treated conservatively. In a randomised trial stenting relieved pain better but caused stent symptoms, so reserve drainage for infection, failing function or genuine obstruction.

3 · Urinary tract infection and drugs

P1 P3 Bacteriuria in pregnancy; Bacteriuria and antibiotic policy in pregnancy — screening culture at 12–16 weeks, 15–45% pyelonephritis if untreated, 4–7 day course, antibiotic safety, ≥14 days for pyelonephritis

Not in the paper answer:

- Non-steroidal anti-inflammatory drugs are contraindicated (third-trimester ductus closure); treat colic with small frequent doses of morphine. Defer intravesical bacillus Calmette–Guérin and mitomycin C until after delivery.
- The **Pregnancy and Lactation Labeling Rule (2015)** abolished the A, B, C, D and X categories in favour of a narrative risk summary.

4 · Imaging

P2 Stones (urolithiasis) in pregnancy — ultrasound first; magnetic resonance imaging (MRI) or low-dose computed tomography (CT) second; cumulative fetal dose ≤50 mGy

Not in the paper answer:

- First-trimester exposure carries teratogenic risk; later exposure mainly raises childhood cancer risk. Needed CT or nuclear studies must not be withheld.
- Give gadolinium only if it clearly improves outcome. Iodinated contrast crosses the placenta but is not teratogenic in animals. A mercaptoacetyltriglycine (MAG3) scan gives a fetal dose

of **<5 mGy**.

5 · Surgery and anaesthesia

- No anaesthetic agent is teratogenic at standard doses. **Never deny or delay necessary surgery in any trimester**, but postpone elective surgery. The **second trimester is safest**.
- Involve obstetrics, monitor the fetal heart, give steroids if the fetus is viable and preterm, and operate where caesarean is available. At laparoscopy, insufflate to **10–15 mmHg**.
- Perinatal loss is driven by the disease, not the operation.

6 · Urolithiasis

P2 P3 Stones (urolithiasis) in pregnancy; Lower ureteric stone <1 cm in pregnancy — calcium phosphate, 50–80% pass, ureteroscopy in any trimester, stent every 4 weeks and nephrostomy every 3 weeks, no shock wave lithotripsy, defer percutaneous nephrolithotomy

Not in the paper answer:

- Nephrostomy carries more preterm labour than a stent. Early ureteroscopy is favoured because repeated exchanges add morbidity and still leave the stone.

7 · Malignancy

Urological cancer is rare (≈ **13 per million** pregnancies) and is diagnosed late.

- **Renal cell carcinoma** is the commonest. Operate as in non-pregnant women, timed to trimester; nephrectomy can be done at caesarean. Avoid targeted therapy and immunotherapy.
- **Angiomyolipoma** is prone to rupture: embolise if stable, operate if unstable.
- **Bladder cancer**: transurethral resection is safe at any gestation; defer intravesical therapy.

8 · Placenta accreta spectrum

Grade	Depth
Accreta	Attached to myometrium
Increta	Invades myometrium
Percreta	Through serosa into bladder, ureters or rectum

- The main risk factor is **previous caesarean**. Blood loss is 3–5 L and maternal mortality up to **7%**.
- MRI maps bladder and ureteric invasion. At cystoscopy, **do not biopsy**.
- Perform **caesarean hysterectomy at ≈ 34 weeks** with the placenta in situ and retrograde ureteric catheters. With detrusor invasion, do a partial cystectomy rather than dissect the plane. Genitourinary injury occurs in ≈ **20–29%**.

9 · Pregnancy after reconstruction

- After augmentation, pyelonephritis occurs in up to **60%**. Use **serum human chorionic gonadotrophin**, because urine tests are falsely positive after contact with bowel mucosa.

Reconstruction	Mode of delivery
Augmentation with intact bladder neck	Vaginal delivery not contraindicated
Bladder neck reconstruction or artificial sphincter	Caesarean preferred by most

Reconstruction	Mode of delivery
Ureterosigmoidostomy	Vaginal delivery risks anal sphincter damage
Bladder exstrophy	Planned caesarean

> **KEY POINT**
> At caesarean after reconstruction, use a **high midline incision** with a urologist present.

- **Myelomeningocele**: latex allergy in up to 60%; folic acid **4–5 mg** daily from 3 months before conception to 12 weeks.
- **Renal transplant**: wait at least 1 year with stable function and stop mycophenolate before conception.

22 Intraoperative Consultations

1 · Principles and ethics

- The patient is unconscious and never consented to the change of plan, so the consultant protects the absent patient without blaming the colleague.
- Accepting the call creates a patient–physician relationship, so disclose the reason fully to the family.
- Four factors decide whether to act now or temporise: patient condition, procedure morbidity, long-term consequences and reversibility.

	Act now	Temporise and defer
Example	Low-grade ureteric injury, treated with a stent	Complex proximal high-grade injury
Morbidity / reversibility	Low / reversible	High / often irreversible
Action	Proceed and disclose afterwards	Await shared decision making

Repair complex injury at once only if deferral is the greater harm. If surgeons disagree, default to the primary surgeon unless that is harmful.

2 · Consultation time-out

ALGORITHM

1. Assess the field first, because the procedure cannot be named until it is seen.
2. Introductions, patient identity, condition, procedure, blood loss, added time and concerns.
3. Verbal agreement by all.

3 · Urethral injury

P1 Outline algorithm for difficult catheter placement — one attempt, then cystoscopy and guidewire, then suprapubic catheter

Not in the paper answer:

- In women, repair a sling or dissection injury vaginally with an interposition flap (peritoneum, omentum or labial fat).

4 · Bladder injury

P2 Lower urinary tract trauma in pelvic fracture — American Association for the Surgery of Trauma (AAST) bladder grades I–V

Not in the paper answer:

- The bladder is the commonest organ injured in pelvic surgery. Retropubic tape injures it in **3–9%** (transobturator 0.5%), so cystoscopy is routine. **Transurethral resection of bladder tumour is the commonest source**: extraperitoneal perforation needs catheter drainage, and intraperitoneal perforation usually needs open repair.
- If injury is suspected but not seen, perform a cystotomy. Close in 1–2 absorbable layers, fill-test, and drain with a urethral catheter for **7–14 days**. No routine suprapubic tube is needed.

KEY POINT

Ask three questions: is it closed, is it drained, and are the ureters intact? At the trigone, when in doubt, reimplant.

5 · Ureteric injury

P3 Ureteric injury: iatrogenic (gynaecological) and upper ureteric injury — AAST grades, retrograde pyelography, repair by level, damage control, prophylactic stents aid recognition only

Not in the paper answer:

- Iatrogenic injury is **75–80%** of ureteric injuries and >90% are in the distal third. Avoid distal ureteroureterostomy (poor blood supply); use reimplant, then psoas hitch, then Boari flap.
- Thermal injury extends beyond the visible zone, so debride generously rather than simply stent.

6 · Renal, bowel and vascular injury

- **PCNL** (percutaneous nephrolithotomy) is the commonest cause of renovascular injury. Venous bleeding is tamponaded by clamping the nephrostomy; arterial bleeding needs selective angioembolisation, not exploration.
- In vascular injury, apply pressure and never clamp blindly.

7 · Placenta percreta and documentation

- The **posterior bladder** is the commonest site of invasion. Never biopsy at cystoscopy (exsanguinating haematuria). Caesarean hysterectomy at **34–38 weeks** is the gold standard; ureters are involved in up to 7%.
- Write a separate note of observed facts even if no injury is found. Never write "careless" or "avoidable".

23 Assessment of Urologic and Surgical Outcomes

1 · Framework

Donabedian	Structure	Process	Outcome
Definition	Context of care	What is done	Effect on patients
Examples	Volume, nurse:bed ratio	Antibiotic and thromboprophylaxis	Mortality, readmission, quality of life
Strength	Easy and cheap	Detects problems early; less affected by case mix	Most meaningful to patients
Weakness	Largely non-modifiable	Valid only if evidence links it to outcome	Confounded, so needs risk adjustment

KEY POINT

Without case-mix adjustment, surgeons who treat high-risk patients look worse, which discourages them from treating such patients.

2 · Long-term end points and bias

P1 Clinical trial methodology; Evidence-based medicine and critical appraisal — lead-time, length-time and Will Rogers bias, Prentice criteria, levels of evidence, type I and II error, number needed to treat

Not in the paper answer:

- **Attribution bias** (cause of death misassigned) is the greatest limitation of disease-specific mortality, because a known cancer is often written on the certificate. **Detection bias** arises when one arm is imaged more.
- The **ICECaP** analysis (Intermediate Clinical End Points in Cancer of the Prostate) found that prostate-specific antigen (PSA) end points did not track overall survival, whereas **metastasis-free survival** did.

P1 RECIST criteria — measurable lesion ≥10 mm, node ≥15 mm short axis, complete and partial response (≥30% vs baseline), progression (≥20% vs nadir and ≥5 mm)

3 · Complications

A complication is any deviation from the expected postoperative course. The Common Terminology Criteria for Adverse Events (CTCAE) are clinician-graded and organ-based, for oncology drugs.

CTCAE grade	Definition
1	Mild; observation only
2	Moderate; minimal intervention; limits **instrumental** activities of daily living
3	Severe; hospitalisation; limits **self-care** activities of daily living
4	Life-threatening; urgent intervention
5	Death

Clavien–Dindo grades by the therapy needed, not perceived severity, which reduces subjectivity.

Grade	Definition	Example
I	No drug, surgical or radiological treatment; antiemetics, analgesics, diuretics allowed; wound opened at bedside	Ileus after cystectomy managed with fluids
II	Other drugs, **including blood transfusion**	Transfusion after nephrectomy
IIIa	Intervention **without** general anaesthesia	Percutaneous drainage of a pelvic collection
IIIb	Intervention **under** general anaesthesia	Unplanned return to theatre
IVa	Life-threatening, intensive care; single-organ dysfunction (including stroke, not transient ischaemic attack)	Myocardial infarction after nephrolithotomy
IVb	Multiorgan dysfunction	Renal failure with respiratory distress
V	Death	—

It does not capture multiple complications, and anaesthesia type varies by institution.

4 · Predicting complication risk

Functional status, comorbidity and frailty predict complications across surgery; adjust for them in every non-randomised comparison.

ECOG	Status	Karnofsky
0	Fully active	100
1	Restricted strenuous activity; light work	80–90
2	Self-care, no work; up >50% of waking hours	60–70
3	Limited self-care; bed or chair >50%	40–50
4	Completely disabled	10–30
5	Dead	0

ECOG is the Eastern Cooperative Oncology Group scale.

- **Comorbidity** drives treatment choice, so it confounds. Administrative data showed higher mortality after transurethral resection of the prostate than open prostatectomy, but adjusting for comorbidity removed the difference.
- **Charlson Comorbidity Index** (predicts 1-year mortality): weight 1 for myocardial infarction, heart failure, vascular, pulmonary disease and diabetes; 2 for hemiplegia, renal disease and any malignancy; 3 for moderate–severe liver disease; **6** for metastatic cancer and AIDS.
- **Fried frailty phenotype**: frail means **≥3 of 5** criteria (weak grip, exhaustion, slow walk, low activity, unintentional weight loss).

5 · Patient-reported outcomes

P1 Health-related quality of life (HRQOL) instruments in urology — generic and disease-specific tools, psychometric properties (Cronbach α >0.7, >0.9 redundant; minimal important difference 3.1 points)

Not in the paper answer:

- Clinicians consistently underestimate urinary, sexual and bowel dysfunction, so collect outcomes directly from patients. A symptom score counts what is wrong; an HRQOL instrument measures function plus bother.

- The International Prostate Symptom Score has 7 items scored **0–35**; the Female Sexual Function Index has **19 items in 6 domains**. Pad tests cannot separate stress from urgency incontinence.

★ Putting it together: everyday decisions

1 · Which imaging test first?

Question	First test
Renal colic	Low-dose non-contrast computed tomography (CT); ultrasound in pregnancy and children
High-risk haematuria	CT urography and cystoscopy
Renal mass	Multiphase CT or magnetic resonance imaging (MRI); enhancement >15–20 Hounsfield units is suspicious
Obstruction, split function	Mercaptoacetyltriglycine (MAG3) diuretic renogram
Prostate cancer before biopsy	Multiparametric MRI
Acute scrotum	Doppler ultrasound, never delaying exploration
Urethral stricture	Retrograde urethrogram

2 · Before theatre

ALGORITHM

1. Assess function, comorbidity and frailty.
2. Stop warfarin 5 days before (international normalised ratio <1.5), bridging only for high thrombotic risk; stop direct oral anticoagulants 48 h before without bridging; withhold metformin on the day.
3. Culture urine before instrumentation; give prophylactic antibiotics.
4. Give extended 28-day low-molecular-weight heparin after pelvic cancer surgery.
5. Complete the World Health Organization checklist.

3 · Irrigation and the infected obstructed kidney

P1 TUR syndrome; Genitourinary infections, bacteraemia and urosepsis — glycine for monopolar resection; drain urgently, stent equals nephrostomy

Not in the paper answer:

- Use saline for bipolar resection, laser and ureteroscopy; never use sterile water for resection, as it causes haemolysis and hyponatraemia.